

Speed of Payment in Procurement Contracts: The Role of Political Connections*

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Abstract

We study a new channel through which politicians exchange favors with campaign donors: faster payment in procurement contracts. We exploit an electoral reform in Brazil that bans corporate contributions and changes the institutional environment in which donor-politician relationships are formed and maintained. Using a within-firm difference-in-differences identification strategy, we find that politically connected firms experience longer average payment terms and a greater likelihood of extreme delays after the reform, especially in cash-constrained municipalities, where payment delays are more prevalent. Our results highlight the importance of designing rules that curb discretion over the contract execution process in government purchases.

Keywords: Payment timeliness, public procurement, political connections

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1 Introduction

Government purchases account for a large share of government expenditures and can be a substantial revenue source for firms (Bosio et al., 2022; Ferraz et al., 2021). To improve efficiency and reduce the scope for corruption, governments have increasingly adopted procurement methods that foster competition and reduce discretion. Yet even under competitive procurement, evidence of *quid pro quo* persists. One possible explanation is that some firms receive preferential treatment after the tendering process, which gives them a competitive edge at the bidding stage.

In this paper, we study a new channel through which politicians may benefit firms that make campaign contributions: more timely payments. Recent research shows that the time elapsed between the delivery of goods and payment is consequential for firms (Abad et al., 2023; Barrot, 2016; Barrot and Nanda, 2020; Breza and Liberman, 2017; Conti et al., 2021). Consequently, the speed of payment can be used by governments to favor certain suppliers. In light of these concerns, several countries have introduced policies aimed at shortening payment terms and limiting discretion over payment practices.¹ Exploring a reform that bans corporate donations, we document an increase in payment time to campaign donors post-reform, particularly in municipalities whose governments are more liquidity-constrained.

Evidence for this channel of political favoritism has been absent in the literature for a variety of reasons. First, its identification requires a shock to the strength of firms' political connections. To address that, we explore a set of electoral reforms in Brazil that change the relationship between donors and politicians. In 2015, corporate donations were banned, and campaign spending limits were imposed. Because these changes occurred in the middle of the mayoral term, firms that donated in the 2012 election cycle were unable to donate again in the 2016 electoral cycle. Even though firm owners could still contribute as individuals, or through illegal channels, the ban coupled with spending limits limited firms' ability to commit to financing political campaigns with the same intensity as before.² If politicians' incentives to grant favors to donors depend on the prospect of raising funds from them in future campaigns, the reforms should be associated with less favoritism.

Second, electoral reforms of this type are not exogenous. They usually coincide with corruption scandals and an increasing anti-corruption sentiment, which can am-

¹Examples include the QuickPay initiative, launched in 2011 in the United States; Regulation 113 of the Public Contracts Regulations, passed in 2015 in the United Kingdom; and Law 21,131 in Chile, enacted in 2019. Brazil's new procurement law (Law 14,133), enacted in 2021, stipulates that payments must be made in chronological order and mandates that all purchasing units publish this order on their websites on a monthly basis.

²Aparicio and Avenancio-León (2022) show that total campaign contributions declined substantially following the ban, although increased individual contributions by affiliates of former corporate donors partially offset this decline.

plify their effects (Clark et al., 2018) or attenuate them if firms and politicians change their behavior before the new rules are in place. Moreover, stakeholders can change their relationship with politically connected firms in response—a bank or a supplier might not want to be linked or financially exposed to a firm that could be charged with corruption and experience distress in the future (Ferraz et al., 2023). In turn, this fact can impair the capacity of politically connected firms to supply to the government and to invest in costly inputs (e.g., labor, legal fees) that enable an effective payment collection process. To account for time-varying firm-level shocks unrelated to the reform, we exploit the fact that firms donated in some municipalities but not others in the last election cycle in which donations were allowed, thereby forming *local* connections. Thus, in municipalities where a firm donated in the previous election, the relationship with the local politician is shaken by the reforms, whereas in municipalities where it did not donate, the relationship remains unchanged. Our setting thus allows us to include firm-by-time fixed effects, which control for firm-level changes and identify the effect from within-firm variation in payment delays around the reform across municipalities with and without local connections.

Finally, data on supplier-level payment delays are often not available to researchers. We leverage a rich new dataset on budget execution at the municipal level in Brazil to construct measures of payment delays at the municipality-supplier level. As is usually the case in budget execution, governments pay suppliers after the verification stage, when they acknowledge that the delivered goods and services comply with contract specifications. We then measure how long governments take to pay suppliers by computing the number of days between the payment and verification dates.

We find that, following the reforms, payment times to connected firms increased by about 2.7 days on average—a modest absolute increase, but a 13% rise relative to the pre-reform average of 20.3 days. We also show that not only does average delay increase, but the likelihood of extreme delays does as well, an arguably more critical concern for firms: the likelihood of payment times exceeding 60 days rises by 3 percentage points, from a baseline of 5%.

We then investigate whether these effects vary across municipalities with different baseline financial conditions. Local governments facing liquidity issues might have to prioritize payments, determining which expenses will be paid on time or with minimal delays. Because firms are not fully compensated for payments that take place outside the contractual terms, late payments can also be interpreted as a partial default, and governments must decide which firms bear the highest losses. We show that the average effects of the reforms on payment times for connected firms are larger, around 6.6 days, in municipalities with lower liquidity. The probability of receiving payment after 60 days increases by 7 percentage points for connected firms in these municipalities, compared to the unconditional probability of 8%. These effects are not statistically

significant in municipalities with higher liquidity.

Another case in which payment timeliness can be relevant occurs when the tendering process is mandatory. In this case, the government’s commitment to paying more quickly may represent an important advantage, allowing favored firms to outbid otherwise similar firms and deterring non-favored firms from participating in or winning procurement contracts (Colonnelli et al., 2024). We then study heterogeneity across competitive and non-competitive procurement methods. We find no statistically significant differential reform effects across procurement methods: the estimates are noisier for both competitive and noncompetitive procurement, ranging from 5–7 days for competitive tenders to 3–5 days for noncompetitive ones. The larger, albeit statistically insignificant, delays for competitive tenders suggest that payment speed may be an important margin of favoritism even when governments find it less straightforward to benefit donors through direct contract allocation. This is especially relevant in our setting, which focuses on “off-the-shelf” products, where auction-rigging is more difficult and costly.

This paper contributes to two broad strands of literature. First, it adds to the substantial literature on the effects of political connections and favoritism (Krueger, 1974; Fisman, 2001; Faccio et al., 2006). The literature has shown *quid pro quo* happening for politically connected firms through a variety of channels, such as preferential procurement contracting (Goldman et al., 2013; Brogaard et al., 2016; Schoenherr, 2019; Baltrunaite, 2020; Gulzar et al., 2025), having more access to finance (Khawaja and Mian, 2005; Li et al., 2008), or beneficial regulation (Fisman and Wang, 2015). Such effects have generally been shown to increase firm value and performance (Fisman, 2001; Jayachandran, 2006; Ferguson and Voth, 2008; Duchin and Sosyura, 2012; Cingano and Pinotti, 2013; Amore and Bennedsen, 2013; Acemoglu et al., 2016; Berkowitz et al., 2021), but at the expense of resource misallocation (Akcigit et al., 2023; Brugués et al., 2024).^{3,4}

We document the timeliness of procurement payments as a novel channel of favoritism, focusing on political donations as an indication of political connectedness.⁵ Our results add to the literature in several ways. First, they offer an explanation for

³In our setting, we are not able to assess whether faster payments to connected firms pre-reform (or the ensuing slower payments post-reform) are welfare improving or not. Since the effects we document are mostly driven by less liquid municipalities, the reform likely changed who receives payment faster—not only among government suppliers, but also staff and other public policies. Making welfare statements would require a broad evaluation of other changes in payment terms across all these actors, as well as measuring effects of delays on affected firms, which are not within the scope of this paper.

⁴An alternative but closely related reading of our findings is that campaign donations privately discipline politicians: once these incentives are removed with the electoral reform, average delay converges to the higher level of non-donor firms, and extreme delays for these donors also increase.

⁵Other papers in the literature define political connections in different ways: CEOs and politicians have educational, professional or social ties, a large shareholder or officer is a member of the parliament or the executive (Bunkanwanicha and Wiwattanakantang, 2008; Tahoun, 2014), a former politician sits on the board of directors (Goldman et al., 2008), among others.

how *quid pro quo* persists even when government agencies use competitive auctions. Second, our measure of favoritism—payment speed—is objective and readily observable. Third, we argue that empirical strategies that use the trajectories of non-donors as a counterfactual for the trajectories of politically connected firms after electoral reforms can be biased due to the endogeneity of the new legislation (e.g., [Besley and Case, 2000](#)). We highlight the necessity of employing within-firm estimations when trying to assess the impact of this type of reform. Finally, our paper sheds light on the nature of donor-politician relationships. The relationship is not based on an instantaneous and one-time exchange of favors; rather, these are relationships built on trust, repeated interactions, and enforced by the future value of complying with the informal arrangement (e.g., [Levin, 2003](#)). Our results show that politicians grant fewer favors to donors (i.e., the relationship is weakened) when they cannot benefit from future donations.⁶

This paper also contributes to a growing literature on the importance of payment terms. [Barrot and Nanda \(2020\)](#) investigate the effects of the QuickPay reform in the US. The reform cut the time to pay from 30 to 15 days for a subset of small firms. They find that treated firms increase employment by 1.7% following the 15-day reduction. [Conti et al. \(2021\)](#) show that the EU directive on late payments increased firm survival, and [Abad et al. \(2023\)](#) document that the payment of government arrears in Spain increased supplier investment. [Barrot \(2016\)](#) shows that stretched payment terms increase barriers to entry and expose firms to liquidity risk. These findings, when applied to this paper’s setting, imply that more favorable payment terms to connected firms affect the ability of non-connected firms to compete in government auctions. Payment terms may be particularly relevant in a developing country like Brazil, where institutional factors, such as weaker enforcement of loan contracts, contribute to increased credit rationing and higher interest rates. The fact that the effect is greater when the municipality has lower liquidity and, therefore, is delaying payments provides an additional motivation for the proper management of government arrears, as their existence might lead to rent-seeking behavior ([Flynn and Pessoa, 2014](#)).

The remainder of the paper proceeds as follows. Section 2 describes the empirical setting, the main data sources, and the empirical strategy. Section 4 presents descriptive statistics. Section 5 discusses the main results, heterogeneity, and placebos. Section 6 concludes.

⁶We also document that these political connections are not likely to be narrowly defined: the effects we document are still present for term-limited mayors, who cannot immediately run for elections. This suggests that donor-politician relationships in these settings are broadly defined, e.g., they could be connections not only to individual politicians but also to parties, or also be forward-looking over the broader career of individual politicians.

2 Setting

2.1 Budget Execution

Two annual laws guide the budget execution of local governments in Brazil: the Budget Guidance Law (*Lei de Diretrizes Orçamentárias*) and the Annual Budget Law (*Lei Orçamentária Anual*). The executive branch proposes the bills, which are discussed, amended, and voted on by the local legislature, and then signed into law by the mayor. The Budget Guidance Law contains the rules that guide the elaboration and execution of the annual budget. It specifies programs that should be prioritized, rules to make budgetary adjustments if realized revenues are smaller than expected, and fiscal targets, including a target for the primary surplus. After the approval of the Budget Guidance Law, the elaboration of the Annual Budget Law commences. The budget details the allocation of expected revenues to each government agency and activity. The budget is comprehensive: an agency can execute an expense only if it is prescribed in the budget.

When the fiscal year starts, the executive branch begins the execution of the expenses specified in the budget. The budget execution process in Brazil is similar to that in other countries (Potter et al., 1999; Flynn and Pessoa, 2014). It consists of three distinct stages:

1. **Commitment:** The government agency reserves part of a specific appropriation for a supplier that was previously selected in a procurement process. From a budgetary perspective, this is the moment an expenditure is recognized since committed amounts are deducted from the budget appropriation.
2. **Verification:** The government agency formally acknowledges that the good or service was delivered according to specifications. This stage requires the issuance and public disclosure of a verification certificate.⁷ This is the moment when a debt to the provider is recognized, and it is considered an expenditure from an accrual-accounting perspective.
3. **Payment:** Cash is transferred to the supplier. This is the moment an expenditure is recognized under cash-basis accounting, which is the method used to compute metrics such as the primary surplus.

The number of days between the verification and payment stages is a proxy for the amount of time it takes governments to pay suppliers. Physical delivery can precede the verification date, especially for products whose verification is more complex and services that do not have a clear delivery date, such as construction. We restrict the analysis to goods to minimize measurement error and because we do not expect the quality of “off-the-shelf” products to decline after the reform (Fazio, 2025). We provide

⁷The certificate is known as a “*nota de liquidação*.” The date on this certificate can be compared with the date on the fiscal receipt signed by the government agency upon delivery.

results for other categories in the Online Appendix.

In 2015, municipalities allocated an average of 12.1% of their total expenses to the purchase of goods. In comparison, expenditures on construction and services provided by firms represented 4.3% and 16.9%, respectively, of total expenses. Salary payments constituted the largest share of municipal expenses, accounting for 54.8% of total expenditures on average.

2.2 Public Procurement and Payments

Government agencies can employ different methods to procure goods and services.⁸ In certain cases, the government can contract directly with a supplier; that is, the tender requirement can be waived. This happens mainly in two cases: when competition is unfeasible (there is only one supplier) or when the purchase is small.⁹ The regulation establishes a threshold to define small. During our sample period, the threshold was 8,000 BRL for products and services and 15,000 BRL for construction. We classify the cases without a tendering process as non-competitive procurement.

The other methods involve a tendering process, such as invitations to tender and reverse auctions (regular and electronic). The method itself depends on the scope and value of the purchase. Even though these methods can differ in some dimensions (e.g., conditions to participate), we classify them in a single group as competitive procurement. The use of electronic reverse auctions has increased over time, especially for simple products.

Government agencies have to pay suppliers within 30 days of acknowledgment that the contract's object has been delivered. When the purchase is small (same thresholds as for direct contracting), the limit is reduced to 5 days.¹⁰ Payments outside the limits are common.¹¹ In such cases, the amount due can be adjusted by inflation and a late payment fee. However, these adjustments are rare and do not fully compensate firms for their losses and increased liquidity risk. Facing delays, suppliers can take the local government to court. However, in addition to being costly, this procedure is unlikely to be effective. Courts are congested in Brazil, and time in court can be long. Suppliers can also decide to terminate the contract, but this decision is only feasible if payment delays are longer than 90 days. In this case, the government is considered to be in default, and delays smaller than 90 days are not considered a contract breach.¹²

⁸During our sample period, the Public Procurement Law (Law 8,666) contained most of the public procurement regulations. The regulation changed in 2021.

⁹The regulation considers other cases, but they are less common. For instance, emergencies and threats to national security.

¹⁰Article 40 of the Public Procurement Law (Law 8,666).

¹¹According to a survey of the National Confederation of Municipalities (*Confederação Nacional dos Municípios*, CNM), 50.2% of the municipalities reported a positive stock of arrears owed to suppliers in 2018 (CNM, 2018).

¹²Article 78 of Law 8,666 describes situations in which the contract can be revoked.

Although the procurement law in effect during our sample period included a provision requiring payments to be made in chronological order, this requirement was not observed. Concerns regarding payment discretion as a potential source of favoritism have been raised in various public debates, especially when governments' financial resources are limited.¹³

2.3 Municipal Elections and Electoral Reforms

Municipal elections occur every four years in Brazil. Voters elect the mayor and city councilors, who serve four-year terms. Elected officials take office on the first of January of the coming year. Mayors may run for re-election but are limited to two consecutive terms. Members of the local council face no term limits and may be re-elected indefinitely. The council is elected in an open-list proportional representation system.¹⁴ Mayors are elected by absolute majority. In municipalities with more than 200,000 voters, there is a run-off if no candidate obtains more than 50% of the votes in the first round. Because of the large number of parties in Brazil, electoral coalitions are common. Among other benefits, coalitions increase the airtime for TV and radio ads, which is free in Brazil and allocated in proportion to the number of seats held by the parties in the federal Congress.

Until 2015, campaigns were financed through private donations and public funds. Individuals and firms could donate to political parties or candidates. Firms could donate up to 2% of their total sales, while individuals could donate up to 10% of their annual income. Candidates, however, faced no limit on self-donations.¹⁵

Starting in 2013, a large anti-corruption investigation revealed a widespread kick-back scheme that involved funding parties with money obtained from federal and state government contracts. Several members of the business and political elite were convicted of corruption charges. Reacting to growing unrest, the judiciary and the federal legislature started to consider measures to deter corruption. In 2013, the Supreme

¹³For instance, the Federal Court of Accounts (*Tribunal de Contas da União*) stated in the report of the Ruling No. 2,360/2018: "Compliance with the payment order is important, since agencies do not always have all the financial resources available. Therefore, a payment order aims to prevent a supplier from being overlooked. The advance of certain payment processes may unduly favor a particular supplier and constitute unequal treatment, thereby amounting to procedural fraud. These advances, when combined with other evidence, may also characterize a crime of corruption." Similarly, [Sarai \(2021\)](#) argues that the chronological order "(...) represents the application of the principle of impartiality in a potential circumstance of finite resources, in which the Administration either does not have the means to pay all of its suppliers, or, if it can do so, runs the risk of having to delay payments to some while the resources for this purpose are determined." In light of these concerns, the new public procurement law, enacted in 2021 (after our sample period), and the Normative Instruction No. 77, of 2022, established that payments should follow the chronological order, which is defined by the verification date, and that government agencies should make the payment order available on their websites.

¹⁴Parties form local coalitions. The number of seats allocated to a coalition is calculated as a proportion of the total number of votes it receives.

¹⁵A limit on self-donations was only introduced in 2019 through Law 13,878/2019

Court began to consider whether the rules allowing campaign contributions were unconstitutional. In September 2015, the Supreme Court declared corporate donations unconstitutional.¹⁶

Also in September 2015, the federal congress passed a law that substantially changed political campaigns.¹⁷ First, it established campaign expenditure limits. The limits were set at 70% of the maximum expenditure observed across all candidates in the previous campaign and adjusted for accumulated inflation between election cycles. Second, the law introduced measures to reduce campaign costs. For example, it cut the campaign period in half, from 90 to 45 days. The rules governing individual donations, whether by candidates or noncandidates, were unchanged.

2.4 Donor–Politician Relationships

The interpretation of our results rests on donor–politician ties being durable and forward-looking, such that the prospect of future donations sustains current favoritism, while the prospect of future favoritism sustains current donations. We provide direct evidence on this premise.

Using data on corporate campaign contributions in the 2008 and 2012 municipal elections (described in Section 3.1), we ask how likely it is for firms to donate consecutively to the same candidate or party, and whether a firm’s supplying relationship with a local government predicts the continuation of its political giving. Conditional on donating in 2008, a firm that sells to the local government is 11 percentage points more likely to donate again in that same municipality in 2012, relative to an unconditional mean of 13%, and this likelihood rises with the amount previously donated (Appendix Table A.1). The tie is also specific: selling to the government raises the probability of donating again to the *same* candidate by 2 percentage points (unconditional mean 5%) and to the *same* candidate or party by 4 percentage points (unconditional mean 8%) (Appendix Table A.2). Among incumbents who won in 2008 and ran again in 2012, where a continued exchange is feasible, these effects rise to 5 and 6 percentage points (unconditional mean 13%) (Appendix Table A.3). In all specifications, the likelihood of repeated donation is also increasing on the amount donated. Appendix Section A

¹⁶The trial started in 2013, and by April 2014, six out of the eleven judges voted against the constitutionality of corporate donations. However, one of the judges requested more time to examine the case, arguing the matter was the prerogative of Congress and not the Supreme Court (see the article “Brazil’s top court bans corporate money in election campaigns,” published at <https://www.reuters.com/article/idUSKCN0RH33A/>). Even though a majority was formed at the beginning of 2014, it was not clear when the court would finish the trial. Cases with vast implications, such as this one, can take many years to be fully appraised, and a judge can single-handedly suspend a case indefinitely. Moreover, as long as the case is open, judges could change their votes. Finally, it was unclear in which elections the new rules would be implemented. It is not uncommon for the Supreme Court to postpone the implementation of a new rule to allow agents to adapt.

¹⁷See Avis et al. (2022) for the effects of this law on political entry and competition.

provides more details.

Although repeat donations are not the norm among all firm-candidate pairs, a commercial relationship with the government substantially raises the likelihood that a firm sustains its political ties over time. This is consistent with the repeated, relational exchange that our identification strategy exploits.

3 Empirical Framework

3.1 Data and Construction of Variables

Our main data source is the MiDES dataset, which harmonizes local budget execution data across states in Brazil (Dahis et al., 2026). We use data collected from the State Audit Courts (*Tribunais de Contas dos Estados*, TCEs) in the states of São Paulo, Minas Gerais, Rio Grande do Sul, and Paraná. These courts are independent institutions that oversee the public finances of municipalities within their respective states. The TCEs of these states provide detailed information on the three stages of budget execution. In particular, they provide the dates and monetary amounts of every commitment, verification, and payment, as well as the supplier’s tax identifier. For municipalities in the state of São Paulo, the data also contain information on the procurement method used by the government to select the supplier.

We select three types of expenses for which the verification date is a good proxy for the delivery date: consumption material, material for free distribution, and equipment and permanent material. We construct two measures: the days between commitment and verification, and the days between verification and payment. The latter is our proxy for the time it takes for the government to pay a transaction. Measurement is straightforward for commitments that have only one verification and one payment. For commitments associated with multiple verification and payment stages, we weigh each operation by its monetary value (see Appendix Figure B.1 for an illustration). Since these are budgetary data, they do not include information on prices and quantities purchased nor details on the tendering process (such as the number of participants or the value of bids).

The Superior Electoral Court (*Tribunal Superior Eleitoral*, TSE) provides political campaign contributions and electoral results data. We collect information for the 2008 and 2012 elections from the Data Basis platform (Dahis et al., 2022). The mayors elected in the 2008 elections were in office from January 1, 2009, to December 31, 2012; mayors elected in the 2012 elections were in office from January 1, 2013, to December 31, 2016 (Appendix Figure B.2 depicts the electoral calendar during our sample period). We observe to which party or candidate the firms donated and in which municipality.

The Ministry of Finance annually provides aggregate data on the financial situation

of municipalities, including balance sheet items, revenues, and expenditures. Balance sheet items are measured on December 31 of each year, while revenue variables refer to the fiscal year, which runs from January 1 to December 31. We construct a measure to assess the liquidity of municipalities. The liquidity measure is defined as the difference between cash and equivalents and a measure of accounts payable, divided by revenues.¹⁸ The higher this measure, the more liquid the municipality, that is, the more liquid reserves a local government has to meet obligations that are due within one year (the current part of the liabilities of a government). We provide more details on the construction of this variable in Appendix Section B.1.

The Brazilian Institute of Geography and Statistics (IBGE) provides data on municipal characteristics, including geographical area, GDP, literacy rates, and population.

3.2 Sample Construction

In this section, we trace all steps we take to arrive at our *full sample* of observations.¹⁹ We start from the universe of budgetary commitments available in the four states and in the two 12-month windows before and after the reform ($N = 39,209,551$).

The budget execution data include all municipal expenses (salaries, pensions, interest payments, machines, equipment, food, office material, construction, IT services, etc.). Brazil adopts a budget classification system in which the economic classification of expenses (*elemento de despesa*) comprises 69 groups identified by a two-digit code. We restrict the data to three groups related to the purchase of goods and materials: consumption material (code 30), material for free distribution (code 32), and equipment and permanent material (code 52). This reduces the sample to $N = 11,250,975$ observations.

We only include fully executed commitments, that is, commitments that are verified and paid within the fiscal year. This reduces the sample to $N = 8,031,841$ observations. We make two final adjustments to our sample of commitments. First, we exclude commitments with payment delays above the 99th percentile of the state-year distribution. Second, as the quality of the data varies across municipalities, we exclude municipality-year pairs in which more than 80% of commitments are verified on the same day of the commitment or paid on the same day of the verification.²⁰ These two adjustments reduce the sample to $N = 6,522,950$ observations, which constitute our main sample of commitments. We then collapse the sample to the firm–municipality–year level, yielding the full sample of $N = 750,621$ observations reported in Table 1, column (1).

¹⁸Similar measures, such as the current ratio and the quick ratio, are used to gauge firms' liquidity.

¹⁹We discuss the data and sample construction in more detail in Appendix Section B.1.

²⁰We show in Appendix Tables C.1–C.3 that using different thresholds does not materially affect our main findings.

3.3 Empirical Strategy

We divide the sample described above into two one-year windows around the electoral reforms of September 2015 (Figure 1a): one year before the reforms (pre-reform) and one year after the reforms (post-reform). The budget execution data are at the commitment level. In our main specification, we collapse them to the firm-municipality-time level using the monetary amount of the commitments as weights. Suppose that in period $t \in \{\text{pre-reform, post-reform}\}$ firm f has C_{fmt} commitments with the government of municipality m . If commitment $c \in \{1, \dots, C_{fmt}\}$ has value V_{cfmt} and days between verification and payment (or days between commitment and verification) D_{cfmt} , then our measure of payment time is

$$y_{fmt} = \frac{\sum_{c=1}^{C_{fmt}} V_{cfmt} \times D_{cfmt}}{\sum_{c=1}^{C_{fmt}} V_{cfmt}}$$

For the amount committed, we simply sum all the commitments of firm f in municipality m and period t .

We run the following regression specification:

$$y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \epsilon_{fmt} \quad (1)$$

where Connected_{fm} is a dummy variable that takes the value one if firm f is connected in municipality m , that is, if it donates to the mayor's party in the 2012 elections; y_{fmt} represents the outcomes of interest (days between commitment and verification, days between verification and payment, and the natural logarithmic of the amount committed), which we measure in periods $t \in \{\text{pre-reform, post-reform}\}$. The dummy Post_t takes the value one when $t = \text{post-reform}$. We control for unobserved time-varying changes at the municipality level, such as a deterioration in the municipality's ability to pay suppliers on time, by including municipality-time fixed effects, α_{mt} . Because the same firm can have contracts in multiple municipalities at a given point in time, we can include firm-time fixed effects, α_{ft} , which absorb firm-level time-varying shocks common across the municipalities in which the firm operates. We also include firm-municipality fixed effects, α_{fm} , which absorb all time-invariant characteristics specific to a given firm-municipality relationship. We cluster standard errors at the firm and municipality levels.

The possibility of controlling for time-varying changes at the firm level is a key advantage of our setting. Electoral reforms of this type often arise in response to corruption scandals, investigations, and anti-corruption sentiment and may therefore coincide with other changes that differentially affect politically connected firms. As a result, connected and non-connected firms may follow different post-reform trends for

reasons unrelated to the electoral reform itself.

The confounds we have in mind operate at the level of the firm. Delayed payment is costly to a supplier regardless of whether it breaches the statutory deadline: the firm must finance the outstanding receivable until cash arrives, which is particularly costly in Brazil because credit-market imperfections lead to credit rationing and high short-term borrowing rates.²¹ A firm's exposure to this cost depends on its overall financial standing as a donor, not on its connection in any single municipality. Suppose, for instance, that banks and other clients limit their financial exposure to politically connected firms. Such a firm faces tighter working capital and a higher cost of carrying any delay, so it may be less able to absorb late payments and could face longer realized payment terms even in the absence of the reform. Alternatively, the same liquidity pressure could push connected firms to intensify collection so as to obtain cash sooner, in which case they might face shorter payment terms even absent the reform. Although these biases could offset one another, it is important to address the possibility that one dominates. Because such firm-level shocks move a firm's financial position and collection ability equally across all the municipalities it serves, the firm-time fixed effects α_{ft} absorb them.

We identify β from variation across municipalities within the same firm and period: for a given firm, we compare municipalities where it donated to the winner with municipalities where it did not. A firm-level confound that shifts delays in connected and unconnected municipalities by the same amount cannot bias β , even when the confound is itself a consequence of the firm being a donor. The identifying assumption is therefore weaker than independence of confounds from connection status: absent the reform, a firm would have followed a similar delay trajectory in its connected and unconnected municipalities.²²

The downside of including firm-time fixed effects is that only firms that transact with more than one municipality enter the sample. Moreover, as we cannot observe the main variables when firms do not sell to the local government, we also restrict the sample to firms that sell in all the time periods of the panel.

Because many firms do not sell to the same local government repeatedly, the inclusion of more pre-reform periods would render these filters excessively restrictive. Moreover, expanding the time window could include the previous or subsequent mayoral term, during which a different mayor may be in office and firms' connection status may change. Therefore, we implement the main analysis with two time periods only

²¹Section 5.6 translates our estimated delays into this financing cost, using the average interest rate on working-capital loans collateralized by receivables.

²²The threat to identification would come from time-varying shocks that affect firms differentially in connected vs. non-connected municipalities, for reasons unrelated to favoritism. For example, α_{ft} does not absorb differential pass-through of a firm-level shock across a firm's own municipalities, such as a donor-firm shock with larger effects precisely where the firm had donated.

(pre- and post-reform) and then perform tests to guarantee that effects are not driven by pre-trends or the specific point in the political cycle in which the reforms occur (around fourteen months before the next election).

To provide evidence in support of the lack of pre-trends, we estimate a similar regression using a sample in which the “post” period is the pre-reform period and the “pre” period is the 12-month window that precedes the pre-reform period (Figure 1b). In this exercise, the definition of connected firms remains the same (a firm is connected if it donates to the mayor’s party in the 2012 elections), but no electoral changes take place between the two periods. In the Online Appendix, we also estimate a dynamic specification with two 12-month pre-treatment periods.²³ To check whether results are driven by the point of the political cycle in which reforms take place, we run a similar regression but using data from the previous mayoral term (four years before). In these regressions, a firm f is connected in municipality m if it donated to the mayor’s party in the 2008 elections. The crucial difference is that firms can donate in the coming elections; that is, no law partially breaks down the relationship between donors and politicians (Figure 1c).

4 Descriptive Statistics

Table 1 provides summary statistics for the main variables. We use two samples: the entire data and a restricted version in which firms are required to sell to more than one municipality.²⁴ The restricted version is our estimation sample, as our identification strategy relies on within-firm variation across municipalities.²⁵ The average time between verification and payment is 18.5 days in the unrestricted sample and 20.3 days in the restricted sample. Despite a low average, Figure 2 shows that a significant share of commitments are paid outside the 30-day period. The 90th percentile is around 42 days in both samples.²⁶

The average time between commitment and verification is 20 days in both samples. The average amount committed is 23,133 BRL in the unrestricted sample and

²³A sample with more than two 12-month pre-treatment periods would contain observations of the previous mayoral term.

²⁴In Appendix Table C.4, we also show the consequences of restricting the sample to firms that sell in both periods. Firms that sell in both periods are larger (in terms of the amount committed) than firms that sell pre-reform only. Appendix Table C.5 further compares the characteristics of firms that sell only before, only after, and in both periods.

²⁵The outcomes of firms that sell to one municipality only are fully explained by the firm-time fixed effects.

²⁶Reliable information on payment times across countries is hard to come by, but municipal governments in Brazil seem to be relatively fast payers. According to data from the Contracting with the Government survey of the World Bank, the average payment time for road construction contracts in other countries was 101 days in 2020. See <https://archive.doingbusiness.org/en/data/exploretopics/contracting-with-the-government>.

35,766 BRL in the restricted sample, which is consistent with firms that sell to more than one municipality having larger contracts. The distribution is highly skewed: the median amount committed is 3,252 BRL in the unrestricted sample and 5,529 BRL in the restricted sample. The averages for the main variables are similar in the pre- and post-reform periods.

Table 2 provides summary statistics of firms and municipalities. We classify firms into three groups. A firm is considered *connected* at a given municipality if, in the previous election, it donated either directly to the elected candidate or to the candidate's party. We divide *unconnected* firms into two groups. An unconnected firm is a *donor* if it donated only to the parties of mayoral candidates that did not win the previous elections, and a *non-donor* if it did not make campaign contributions. Unconnected non-donor firms comprise the majority of our sample. Connected firms have larger contracts (as measured by the amount committed) than unconnected donors, which in turn have larger contracts than unconnected non-donor firms. There are several potential explanations for this pattern. Firms that donate could be more efficient and have a larger production capacity. Additionally, connected firms can engage in activities that are part of the incumbent's agenda. For instance, a firm that produces school supplies might donate to candidates who prioritize education expenditures. The allocation of contracts to firms that have a close relationship with incumbents can also be a solution to issues such as moral hazard or adverse selection when the quality of the object of the contract is not readily observable or verifiable. Finally, donors could have larger contracts because of favoritism.²⁷

Even though connected firms sell to 13 municipalities on average, they only donate to 1.2 municipalities on average. The amount connected firms donate to winners is around 10,123 BRL, which represents 1% of the average amount committed to these firms. The time between verification and payment for connected firms is about one to two days shorter than for unconnected firms.

In Panel B of Table 2, we show that the average municipality in our sample has 36 thousand inhabitants and a GDP per capita of 22.4 thousand BRL in 2015. We split the sample of municipalities into high- and low-liquidity groups using the median of the liquidity measure in 2015 within states. High- and low-liquidity municipalities display similar levels of GDP per capita and population. However, low-liquidity municipalities pay suppliers 5.5 days later than high-liquidity municipalities. In Figure 3, we plot the distribution of days between verification and payment for the two groups of municipalities. The share of commitments paid after 30 days is larger in low-liquidity mu-

²⁷In Appendix Table C.6, we show that not only connected firms receive larger committed amounts, but they receive more in precisely those municipalities where they are connected, i.e., where they donated to the winning mayor in the 2012 election. This is also true within firms: for the same firm, the amount committed in the data is 1.2 log-points higher in municipalities where it is connected than in municipalities where it is not.

municipalities, confirming that delays are more common when governments have fewer liquid resources available to meet short-term liabilities. Appendix Figure C.1 shows how our payment time measure correlates with business cycle fluctuations. It shows that payment times increase during recessions. Moreover, low-liquidity municipalities take longer to pay suppliers during both expansions and recessions.

5 Results

5.1 Main Results

Table 3 reports the coefficients obtained from the estimation of Equation 1. We report results in five combinations of fixed effects. First, we include no fixed effects to allow for the full variation in the data, with firms that sold to governments before or after the reform, or both. Second, we include municipality-time fixed effects to account for municipality-specific trends. Third, we include firm-time fixed effects to account for firm-specific trends. Fourth, we include both sets of fixed effects along with firm-municipality fixed effects, to estimate our most stringent specification discussed in Section 3.3. In Appendix Table C.7 we report an alternative in which all columns are restricted to the common sample that survives this most stringent specification, with similar results.

The time between verification and payment for connected firms increases by approximately 4 days after the reforms, once we include firm-time fixed effects, exploring within-firm variation in political connection. The magnitude of the coefficient reduces to 2.8 days and 2.7 days when we include municipality-time FE (column 4) and firm-municipality FE (column 5), but remain statistically significant at 5%. This magnitude is around 13% of the average time between verification and payment variable (20.3 days). In Section 5.6, we provide a more detailed discussion about the economic significance of the effect. In Figure 4, we show that the difference-in-differences effect comes from connected firms being paid earlier than unconnected firms before the reforms; after the reforms, the two groups face similar payment times.²⁸

Even though we restrict the sample to simple products with clear delivery dates and relatively straightforward verification, we examine the possibility that connected

²⁸Results are robust to a broader definition of political connection based on donations to any party in the winning mayor's coalition rather than only the mayor's party (Appendix Tables C.8 and C.9).

firms may still benefit from more timely verification.²⁹ Panel B of Table 3 examines the number of days between commitment and verification and shows no change in this measure for connected firms after the reform: the estimated coefficients are an order of magnitude smaller and not statistically different from zero.

For non-competitive procurement methods, the allocation of a contract is arguably the first-order channel through which politicians can favor connected firms. The breakdown of the relationship between donors and politicians would be followed by a smaller amount committed. In competitive procurement, the government's commitment to pay earlier enables connected firms to outbid non-connected firms that are otherwise similar. Therefore, the amount committed and payment timeliness are jointly determined, and a deterioration in payment timeliness would also be followed by a decrease in the amount committed. Panel C of Table 3 shows no effects on the log of the amount committed for connected firms after the reforms. One potential explanation for the absence of an effect is that the amount committed variable displays more inertia than the time variables. The length of the contracts can be as long as five years, especially for large amounts, and commitments after the reform could refer to contracts awarded before the reform.

A potential concern with our main specification is that aggregating commitments to the firm-municipality-time level could mask changes in the composition of contracts, for example, if connected firms shift toward contract types that are intrinsically paid faster. We address this by re-estimating the effect directly at the commitment level for the state of São Paulo, the only state for which we observe the expenditure category and the procurement method of each commitment³⁰. Without composition controls (other than committed amounts), the commitment-level estimate is 3.5 days; adding sub-category-by-time and procurement-method-by-time fixed effects leaves it essentially unchanged, at 3.6 days (Appendix Table C.12). The effect is therefore not driven by connected and unconnected firms holding systematically different types of contracts.³¹

²⁹Alternatively, repeated donor-politician interactions may mitigate adverse selection and moral hazard—for example, concerns about the delivery of low-quality products—allowing governments to spend less time assessing the quality of products supplied by connected firms. If the reform disrupts these relationships, connected firms may have weaker incentives to maintain product quality, requiring governments to devote more time to inspection before verification. Breza and Liberman (2017) show that buyers use delayed payment terms to assess product quality. The idea that delayed payments can mitigate concerns about product quality dates back to Smith (1987), Lee and Stowe (1993), and Long et al. (1993).

³⁰Results for our main specification in the state of São Paulo, including with heterogeneous effects by liquidity levels, are presented in Appendix Tables C.10 and C.11.

³¹Another possible concern is that contracts of different sizes are systematically paid differently, and our results could partially capture differential impacts of the reform in large vs. small contracts. Other than directly controlling for contract size, in Appendix Table C.13 we also show that, pre-reform, commitments above and below the median amount faced similar average delays and were also awarded to connected firms at similar rates.

5.2 Heterogeneity by Local Government Liquidity Status

In Table 4, we show how the municipality's liquidity shapes results. We divide the sample into two groups of municipalities (high- and low-liquidity) using the median of the within-state liquidity measure in December 2015.³² In our most stringent specifications, the time between verification and payment for connected firms increases by 6.6-7.2 days after the reform in low-liquidity municipalities.³³

This result is consistent with payment timeliness being a more relevant dimension of favoritism when municipalities experience difficulties in meeting short-term obligations. In such cases, governments must decide which expenses are paid according to contract terms and which may experience smaller delays. Since the electoral reforms reduce the value of relationships with connected firms, cash-constrained governments may shift their priorities, allocating funds to other expenses that are now comparatively more valuable.³⁴ If liquidity shocks to the municipality government coincide with liquidity shocks to the firms that trade with it, this type of favoritism is even more relevant as it takes place when the marginal value of cash for the firms is high. The favoritism, in this case, would have an insurance aspect: it pays off exactly when its marginal value is higher. Payment timeliness is less of a problem when governments have enough cash to pay all suppliers on time. Indeed, in high-liquidity municipalities, point estimates are around zero and not statistically significant.

5.3 Extreme Delays

It could be that not only does the average payment speed change, but extreme delays also become more likely. Extreme delays may be a primary concern for suppliers, especially if delay costs increase exponentially over time. We assess whether, after the reforms, municipalities are more likely to make payments to connected suppliers 30,

³²Even though the number of municipalities is the same in both groups, the number of observations is higher in the high-liquidity sample. This is because the data are at the firm-municipality-time level, and high-liquidity municipalities are slightly larger and richer than low-liquidity municipalities (Table 2).

³³Throughout the heterogeneity and robustness analyses, the tables in the main text report only the days between verification and payment outcome, which corresponds to Panel A of Table 3; the corresponding estimates for the days between commitment and verification and for the log of the amount committed are reported in the Online Appendix. The effects on the other variables—days between commitment and verification and log of the amount committed—are not statistically significant in either subsample.

³⁴Firms might also understand that, pre-reform, liquidity constraints make donations more valuable if they are an insurance against late payments in dire times. In Appendix Table C.14, we examine whether municipality liquidity and other characteristics predict a firm's decision to donate, exploiting variation within firms. Municipality population is the only significant correlate of the decision to donate in both our full and estimation samples, while donation is marginally less likely in municipalities with rigid budgets in our estimation sample. Therefore, we do not find that firms are particularly likely to donate in low-liquidity municipalities, but acknowledge that the donation decision is observed in equilibrium and firms are likely to also donate selectively.

45, or 60 days after the verification stage. We provide descriptive statistics of these variables in Appendix Table C.15. In the estimation sample, the probabilities of being paid 30, 45, and 60 days after the verification stage are 21%, 10%, and 5%, respectively. The table also reveals that extreme delays are more common in liquidity-constrained municipalities, where the probabilities of payment times exceeding 30, 45, and 60 days rise to 27%, 14%, and 8%, respectively.

In Table 5, we show that, after the reforms, the probability of being paid in more than 60 days increases by 3 percentage points for connected firms, compared to an unconditional mean of 5%. The probabilities of being paid in more than 45 or 30 days also increase by 3 percentage points (relative to the unconditional probabilities of 10% and 21%, respectively), but we cannot reject the hypothesis that these effects are different from zero.

In Table 6, we split the sample by the liquidity status of the local government (as in the previous section) and find that the effects for connected firms that sell to illiquid municipalities are much larger. In illiquid municipalities, the probability of being paid after 45 days increases by 10 percentage points for connected firms (compared to an unconditional mean of 14%), and the probability of being paid after 60 days increases by 7 percentage points (relative to an unconditional mean of 8%). We find no statistically significant effects for municipalities with high liquidity.

5.4 Heterogeneity by Competitive and Noncompetitive Procurement

In Table 7, we test whether effects differ by the type of procurement method used. We restrict the data to the municipalities of the state of São Paulo since this information is not available in other states. Competitive procurement methods entail an open tendering process. Non-competitive procurement methods do not involve a tendering process and, as a result, politicians have more discretion in the selection of suppliers.

In our preferred specifications, we cannot reject that effects are equal in both types of procurement methods: we estimate an increase in payment delays in the range of 5–7 days for competitive methods and 3–5 days for non-competitive ones, although estimates are noisier in these smaller samples. The finding that the political connection channel we document is also present in competitive contracts, with a magnitude similar to that in non-competitive contracts, suggests that speed of payment is an important way of distorting public procurement even when it is more difficult to award contracts to connected firms. We do not find statistically significant effects in the other variables (days between commitment and verification and log of the amount committed). In the case of the amount committed, one possible explanation is the unobserved length of the contracts, which means that commitments after the reform could refer to contracts signed before the change. Although competitive contracts are substantially larger in

committed value than non-competitive ones, the two groups display similar average times between verification and payment before the reform (Appendix Table C.16), so the heterogeneity by procurement method is not a proxy for contract size.

5.5 Placebo Tests and Pre-Trends

In Table 8, we report the results of placebo exercises with two periods. A potential concern is that, since the post-reform period overlaps with the 12-month period before the next election, results could be driven by politically connected firms allowing governments to prioritize the payment of other expenses on time. For instance, as payment times of some expenditures (such as the salaries of school teachers) are very costly politically and can affect election results, connected firms might put up with delays so that governments can pay these expenses on time. To rule out this channel, we run the same specification at the same point of the mayoral term (that is, fourteen months before the next election) but in the previous electoral cycle, when reforms did not take place (Figure 1c). Estimates in columns (1) through (5) are not significant and small in magnitude, providing evidence that the effects we uncover in Table 3 are not driven by features of the political cycle.

A possible concern with our pre-post differences-in-differences approach is the existence of pre-trends or anticipation effects. Because we exploit variation at the firm-municipality level and not all firms will receive commitments in municipalities in all periods, using higher-frequency data means that our sample of firm-municipality pairs changes substantially across periods, resulting in a highly unbalanced panel and making the estimates from a dynamic difference-in-differences specification more difficult to interpret.

We provide several pieces of evidence that reduce concerns with differential pre-trends and anticipation effects, sequentially allowing for higher-frequency data while acknowledging the associated limitations. First, in columns (6)-(10) of Table 8, we run the same specification as our main results, but defining the pre-reform period (August/2014-July/2015) as the “post” period, while the 12 months before that (June/2013 - May/2014) are defined as the pre-period. Coefficients of interest are smaller and not statistically different from zero, suggesting no differential trend in these two 12-month periods. We then report a similar specification of a dynamic difference-in-differences model with four 12-month periods: two periods before the reform (2013 and 2014), the reform year (2015), and one year after the reform (2016) (Figure 5 and Table 9). In our preferred specification with firm-municipality fixed effects, we find no significant differential trends for connected pairs in 2013 and 2014 relative to the 2015 baseline, followed by a significant 3.3-day increase in payment delays post-reform in 2016, similar to the effect estimated in our main specification.

Finally, in Appendix Figure C.2, we compute monthly average delays for firm–municipality pairs between January 2014 and September 2016, separately for connected and unconnected pairs.³⁵ We find that, consistent with other evidence, firms received faster payment in connected municipalities pre-reform; post-reform, the average delay for connected pairs increases and narrows the gap with that of unconnected pairs.³⁶

We also acknowledge the threat of anticipation: since the judicial case that eventually banned corporate donations went on for years, it is possible that firms and politicians changed their behavior in anticipation of the final decision in September 2015. If this was indeed the case, that would bias our estimates towards zero, as no differential impact in payment delays would be observed in the months following the reform. But we believe anticipation is unlikely to be a major concern in our setting. First, as discussed in Section 2, there was uncertainty about when the Supreme Court of Brazil would make a final decision. Moreover, major decisions of this type are often implemented with a delay, allowing agents time to adjust. Thus, even if the decision itself was partly anticipated, its immediate application to the upcoming elections likely involved a substantial degree of surprise. Second, despite the caveats discussed above, our higher-frequency estimates do not provide evidence in favor of anticipation behavior, which would show up as differential pre-trends.

5.6 Economic Significance and Factors That Can Attenuate the Effects

What is the economic significance of the results? Estimates of the effect of the reforms on the time between verification and payment range from 2.7 in the main sample to 7 days for low-liquidity municipalities. Barrot and Nanda (2020) find that a 15-day reduction in payment terms in the United States causes an increase of 1.7% in firms' employment. In Brazil, because financial frictions are more severe and access to credit is more restricted and expensive, a similar reduction in payment times could have even larger effects. However, the effects do not seem to be large in monetary amounts. Assuming that firms finance their working capital needs with bank loans that use receiv-

³⁵We construct the underlying sample from the firm–municipality pairs that remain in the baseline difference-in-differences sample after including municipality-time, firm-time, and firm–municipality fixed effects.

³⁶In Appendix Figure C.3, we also estimate a formal dynamic DID specification, presenting the coefficients for the interaction with the indicator for political connection and month, omitting August 2015 as reference pre-reform month. We present two specifications: one with firm-month and municipality-month fixed effects, and another that additionally includes firm-municipality fixed effects. Results are not very precise, but in both specifications pre-reform coefficients are distributed around zero. In the first specification, we also find a significant increase in payment delays for connected pairs, particularly six months after the reform. As previously discussed, the main challenge to interpreting these coefficients is the unbalanced nature of our panel: in Appendix Table C.17, we show the number of observations in each month and document that it varies significantly, from less than 14,000 pairs to more than 60,000 in the first specification. This variation arises because firms do not receive commitments every month. Given the resulting changes in sample composition across months, we place less weight on these estimates.

ables as collateral, 2.7-7 days represent around 0.3-0.6% of the amount committed.³⁷

Other institutional features may moderate the magnitude of these effects. Because firm owners can still donate as individuals, the breakdown of firm–politician relationships is likely to be only partial. Indeed, [Aparicio and Avenancio-León \(2022\)](#) show that increased individual contributions from affiliates of former corporate donors partially compensated for the loss of corporate donations. Donors may also continue contributing illegally. In Brazil, such contributions are common and often take the form of unreported slush funds used by politicians to finance their campaigns. It is therefore difficult to assess the extent to which the reforms severed the relationships between firms and politicians. Nevertheless, the estimated magnitudes can be interpreted as lower bounds on the effects that would arise from a complete breakdown of these relationships.³⁸

Moreover, in this paper, we focus on simple products. Firms selling these products likely operate at low margins, so the effect can be substantial relative to their margins. Moreover, relatively few of these firms donate: only 21% of donations in the 2012 elections came from firms in our sample that contracted with municipalities throughout the mayoral term (2013–2016). Most donations come from firms in other sectors, particularly construction. One possible explanation is that procurement may be easier to manipulate in construction, where supplier selection can depend on technical capabilities and other dimensions in addition to price. However, favoritism through discretion over the stages of budget execution may still be important. Because completion is more difficult to verify for construction projects and there is often no single delivery date, agencies have an additional margin of discretion: the verification stage. By postponing certification that the contract has been executed according to specifications, agencies can delay payment, so discretion over both verification and payment may increase the scope for favoritism through payment terms. A similar argument applies to services delivered continuously rather than on a single date. Although we find no statistically significant effects for services and construction in Appendix Table [C.21](#), these estimates should be interpreted with caution because payment speed is more prone to measurement error.

³⁷According to the Central Bank of Brazil, the average monthly interest rate of loans that use receivables as collateral was 2.31% in December 2015.

³⁸Term limits provide another potential source of attenuation. Brazilian mayors may serve at most two consecutive terms, so second-term mayors cannot run for reelection and may have weaker incentives to favor previous donors. We test this hypothesis using a triple-difference specification in Appendix Table [C.18](#) and find no evidence that our effects differ for term-limited mayors. We complement this with a regression discontinuity design comparing municipalities where an incumbent narrowly won reelection with those where a challenger narrowly won a first term (Appendix Table [C.19](#)), and with a placebo of the same exercise in the previous electoral cycle (Appendix Table [C.20](#)); both are imprecise given the smaller samples but consistent with this conclusion. A possible explanation for those findings is that political connections are not only specific to a candidate in the upcoming election: if firms are connected to political parties, or expect to have a relationship over future elections with the mayor ([Snyder, 1992](#)), then term-limits might not be a primary driver of the results we document.

6 Conclusion

This paper provides evidence that payment terms to campaign donors change after an electoral reform that bans corporate political contributions. The reform limits the ability of firms that donated in previous elections to fund future campaigns, partially breaking down their relationships with politicians. The changes are more pronounced in municipalities with lower liquidity, where resources available to meet short-term obligations are scarcer. Our findings highlight a previously overlooked channel through which politicians can grant favors to donors. Moreover, this channel can distort the allocation of public contracts even under mandatory competitive auctions. Preferential payment terms may weaken the competitiveness of non-connected firms, particularly those facing tighter financial constraints, and thereby help explain why connected firms are more likely to win competitive auctions.³⁹

The paper also sheds light on the informal relational contract between politicians and donors. In particular, it highlights the fact that the prospect of receiving future donations is a key incentive for politicians to grant favors. From a policy perspective, the results call for rules that curb discretion over payment dates and properly compensate firms for late payments.⁴⁰ Moreover, the results being larger in municipalities with low liquidity and larger payment times provide an additional motive for policies aimed at preventing the build-up of government expenditure arrears, as chronic delays might incentivize the rent-seeking behavior we uncover in this paper.

³⁹In this paper, post-bidding preferential treatment takes the form of shorter payment delays. Other forms of post-bidding favoritism are also possible. Politicians may, for example, reduce execution costs by requiring less paperwork or providing greater administrative flexibility. In settings with substantial uncertainty over execution costs, such as infrastructure projects, renegotiations are common, and politicians may also offer more favorable renegotiation terms (Brogaard et al., 2021).

⁴⁰As an example, the 2021 procurement law reform in Brazil established that payments should be settled on a first-come-first-served basis.

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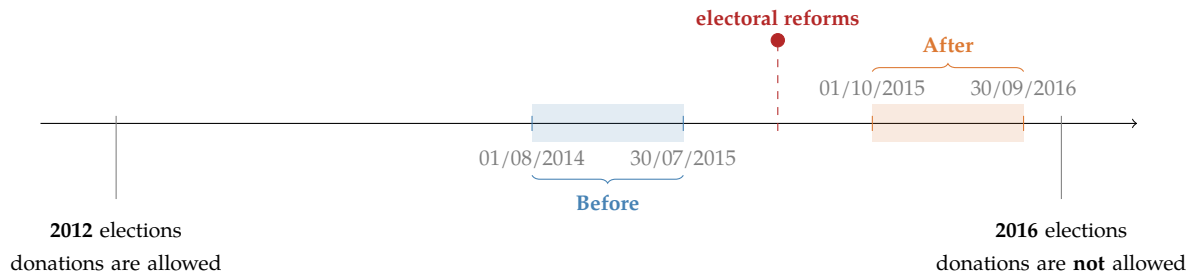
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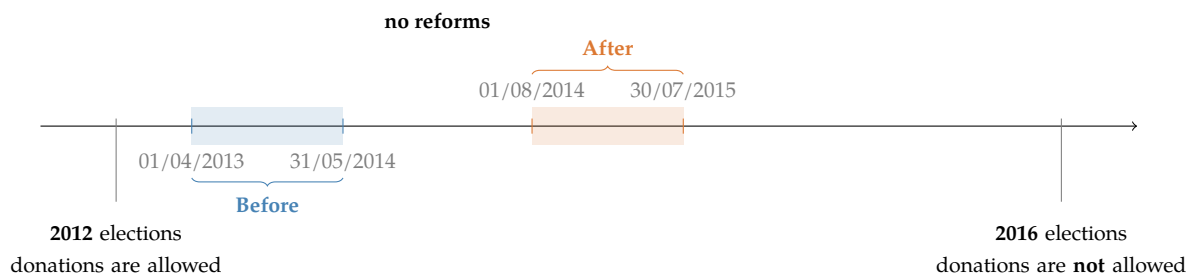
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Figures

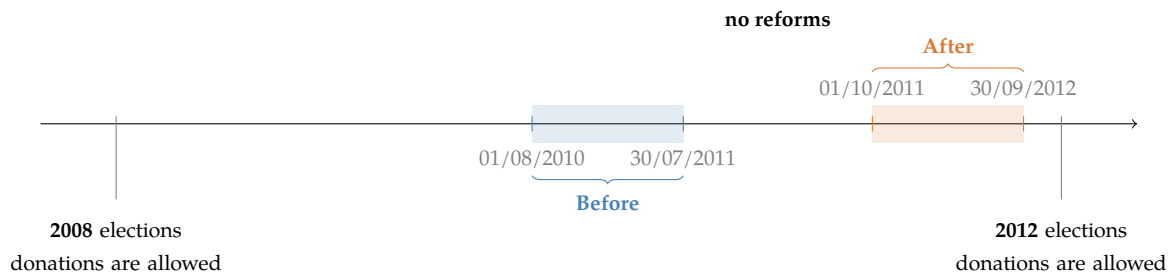
Figure 1: Estimation strategies



(a) **Main sample:** estimation around the 2015 electoral reforms



(b) **Pre-trend:** estimation before the 2015 electoral reforms, within the same electoral cycle

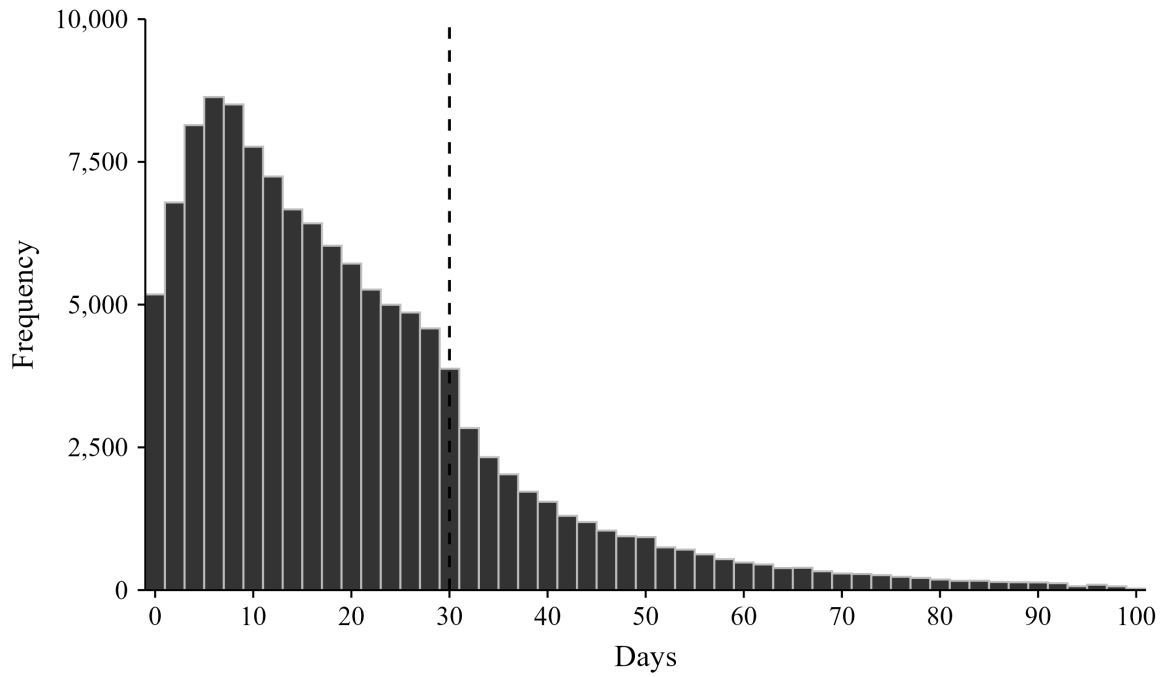


(c) **Placebo:** estimation in the previous electoral cycle

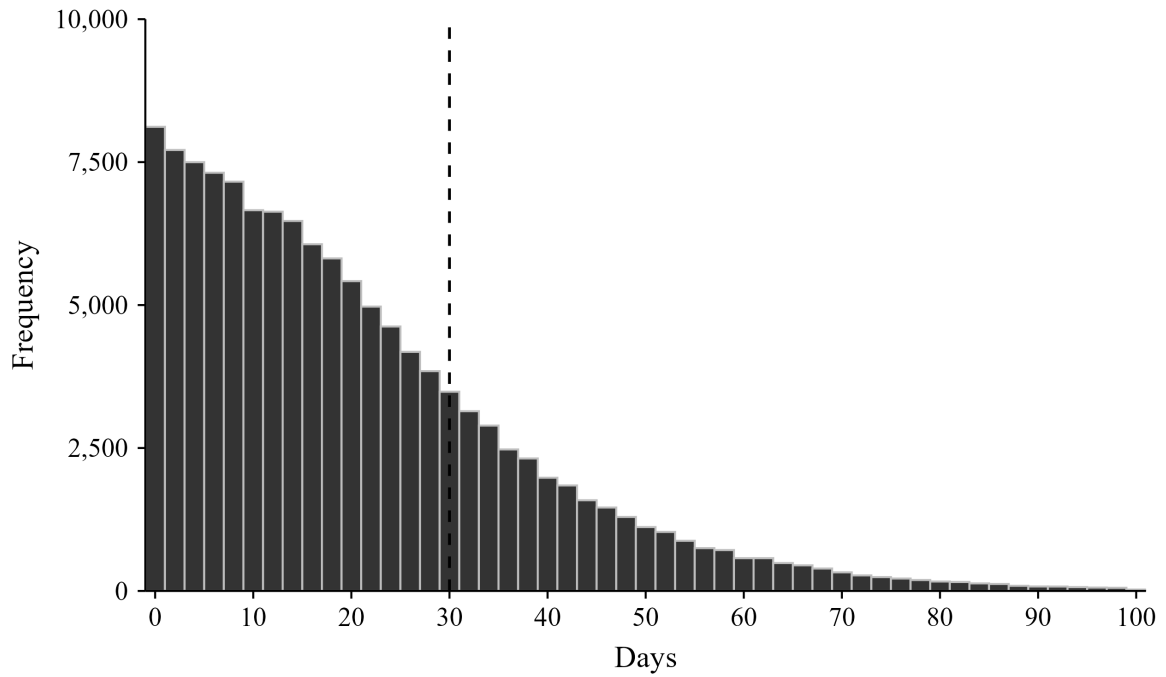
■ Before window ■ After window

Notes: The figure illustrates the construction of the datasets used in the main analysis (Panel (a)), a pre-trend exercise (Panel (b)), and a placebo exercise (Panel (c)). The main analysis compares observations before and after the reforms. The pre-trend exercise treats the original pre-reform window as the post period and compares it with a pre period obtained by shifting the original pre-reform window back by one year. The placebo exercise compares the same points in the electoral cycle as in the main analysis, but in the previous electoral cycle, during which no reform took place.

Figure 2: Histograms



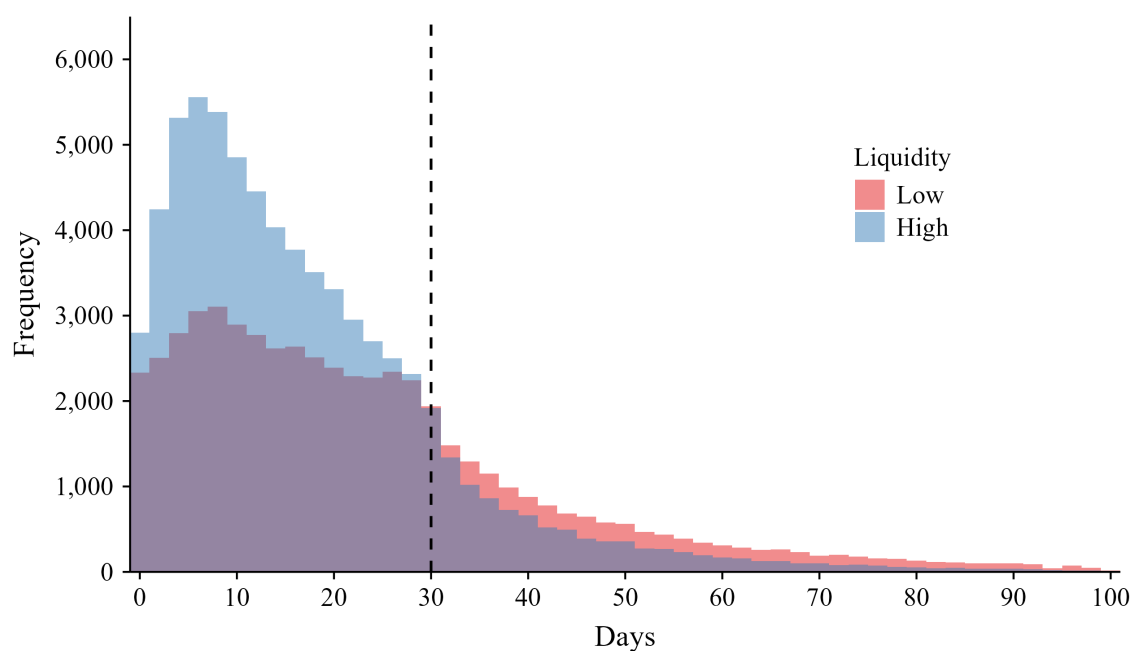
(a) Days between verification and payment



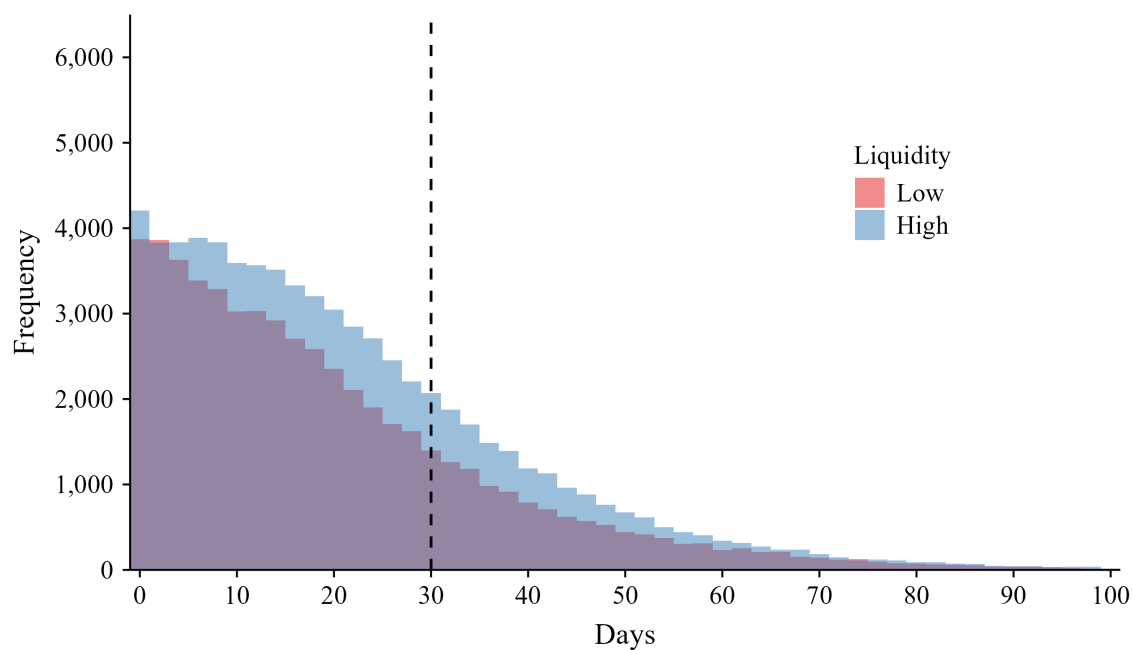
(b) Days between commitment and verification

Notes: Each observation is a firm-municipality pair. The variables are commitment-value-weighted averages calculated over the pre- and post-reform times. The sample is the baseline estimation sample retained by the fully saturated difference-in-differences specification, which includes municipality-time, firm-time, and firm-municipality fixed effects. Histograms use two-day bins.

Figure 3: Histograms: heterogeneity by the liquidity of local government



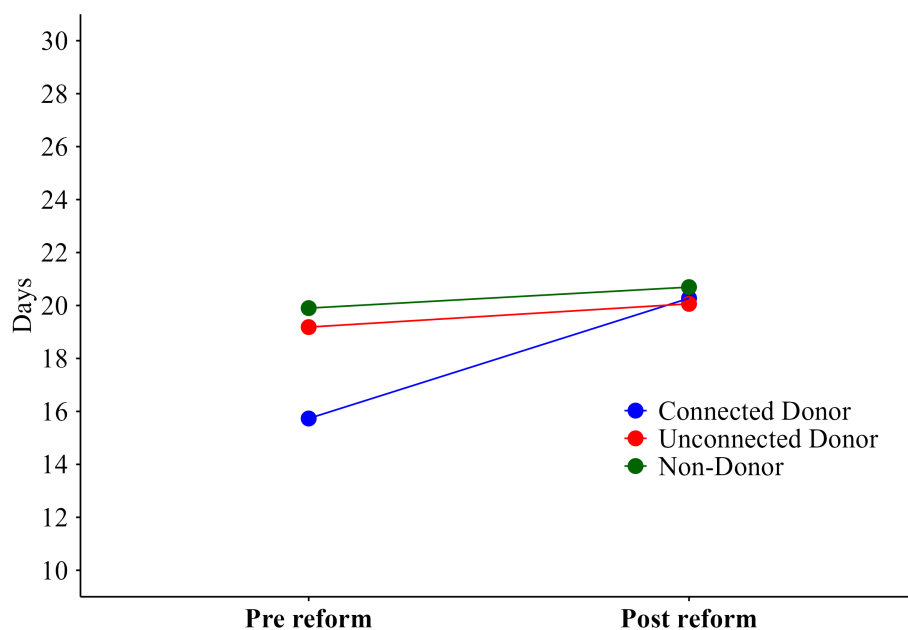
(a) Days between verification and payment



(b) Days between commitment and verification

Notes: Each observation is a firm-municipality pair. The variables are commitment-value-weighted averages calculated over the pre- and post-reform times. The sample is the baseline estimation sample retained by the fully saturated difference-in-differences specification, which includes municipality-time, firm-time, and firm-municipality fixed effects. Municipalities with low (high) liquidity are those for which the liquidity measure in December 2015 is below (above) the median. Histograms use two-day bins.

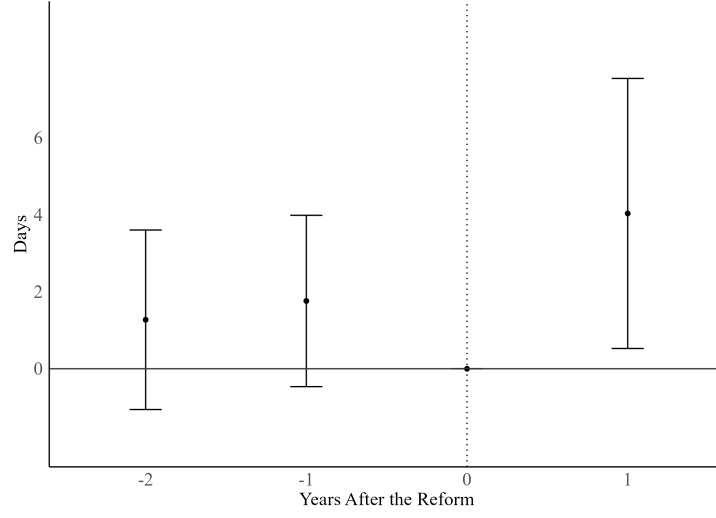
Figure 4: Days between verification and payment averages for different groups, before and after the reform



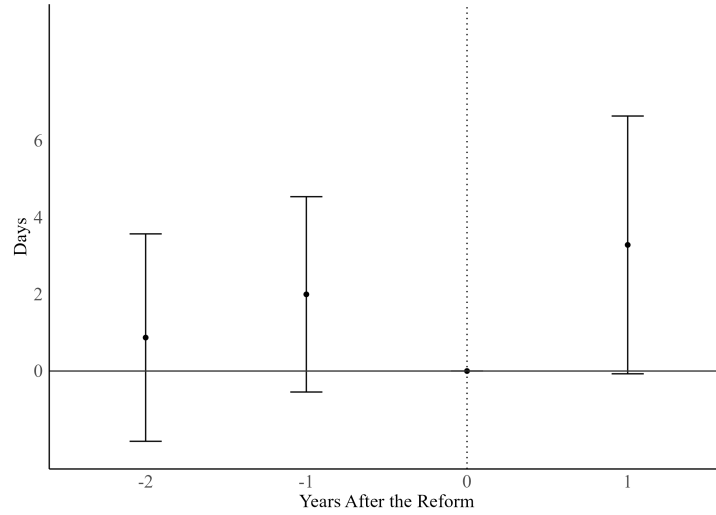
Notes: The figure reports pre- and post-reform averages of the days between verification and payment variable. The sample corresponds to the one used in the main difference-in-differences specification with municipality-time, firm-time, and firm-municipality fixed effects. Connected firms are those that donated to the winning mayor's party in the 2012 election; unconnected-donor firms are those that donated to parties of candidates who were not elected in the 2012 election; unconnected non-donor firms are those that did not make campaign contributions.

Figure 5: Days between verification and payment: pre-trends

(a) Firm-time and municipality-time fixed effects



(b) Adding firm-municipality fixed effects



Notes: The data are at the firm–municipality–year level. The figures report coefficients from event-study regressions of days between verification and payment on interactions between firm–municipality political connection status and year indicators, omitting 2015 as the reference year. Panel (a) estimates $y_{fmt} = \beta_{-2} \text{Connected}_{fm} \times I_{t=2013} + \beta_{-1} \text{Connected}_{fm} \times I_{t=2014} + \beta_1 \text{Connected}_{fm} \times I_{t=2016} + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \epsilon_{fmt}$. Panel (b) adds firm–municipality fixed effects, α_{fm} . Connected_{fm} equals one if firm f was connected to the winning mayor’s party in municipality m in the 2012 election. $I_{t=2013}$, $I_{t=2014}$, and $I_{t=2016}$ are year indicators. Standard errors are two-way clustered at the firm and municipality levels.

Tables

Table 1: Descriptive statistics of main variables

		(1)	(2)	(3)	(4)
		Full sample	Estimation sample	Estimation sample - pre	Estimation sample - post
Amount committed (BRL)	Mean	23,133	35,767	35,863	35,670
	SD	149,524	206,989	218,618	194,667
	p10	269	480	500	458
	p50	3,252	5,529	5,708	5,357
	p90	40,087	64,709	64,956	64,459
Days between commitment and verification	Mean	20.2	20.4	20.5	20.4
	SD	21.1	19.6	19.6	19.5
	p10	0.1	1.0	1.0	1.0
	p50	14.0	15.1	15.2	15.0
	p90	48.7	46.6	46.8	46.3
Days between verification and payment	Mean	18.5	20.3	19.9	20.7
	SD	20.7	20.4	19.9	21.0
	p10	1.0	1.9	1.7	2.0
	p50	12.4	15.0	14.8	15.0
	p90	41.9	43.7	42.7	44.8
Observations		750,621	248,808	124,404	124,404

Notes: The table presents descriptive statistics of the main variables: mean, standard deviation (SD), 10th percentile (p10), median (p50), and 90th percentile (p90). The data are at the firm-municipality-time level. The pre-reform period is the 12-month window before the reform (August 1, 2014–July 30, 2015), and the post-reform period is the 12-month window after the reform (October 1, 2015–September 30, 2016). Column (1) uses the sample without fixed effects. Column (2) uses the estimation sample with firm-time, municipality-time, and firm-municipality fixed effects. Columns (3) and (4) use the sample in column (2), restricted to the pre-reform and post-reform periods, respectively.

Table 2: Descriptive statistics for firms and municipalities

	(1)	(2)	(3)
		Unconnected	
<i>Panel A: Firms</i>	Connected	Donor	Non-donor
Days between commitment and payment	39.2 (25.6)	40.8 (26.2)	40.7 (27.7)
Days between commitment and verification	21.2 (19.1)	21.2 (18.6)	20.4 (19.6)
Days between verification and payment	18 (17.7)	19.6 (19.9)	20.3 (20.5)
Number of municipalities firm sells to	13.3 (39.1)	10.2 (24.6)	5.4 (12.7)
Total committed (1,000 BRL)	1,020 (3,953)	471 (1,572)	183 (1,112)
Number of municipalities firm donates in	1.2 (0.6)	1.1 (0.3)	- -
Total donated to losers (BRL)	2,881 (14,587)	8,580 (31,299)	- -
Total donated to winners (BRL)	10,110 (19,819)	- -	- -
Number of connections	1.1 (0.4)	- -	- -
Number of firms	346	354	21,585
		Liquidity	
<i>Panel B: Municipalities</i>	All	Low	High
Liquidity, 2015	0.06 (0.17)	-0.02 (0.07)	0.17 (0.17)
GDP per capita (1,000 BRL), 2015	22.4 (20.6)	23.4 (19.2)	25.8 (24)
Population (1,000)	36.1 (251.2)	34.5 (126.5)	36.4 (93)
Days between commitment and verification	19.1 (10.6)	18.1 (10.4)	20.1 (10.7)
Days between verification and payment	21 (11)	23.8 (11.1)	18.3 (10.2)
Total committed (1,000 BRL)	4,347 (11,849)	3,796 (12,143)	4,896 (11,605)
Number of municipalities	2778	996	1037

Notes: This table presents the mean and standard deviation (in parentheses) of firm and municipality characteristics. The firms are those that enter our estimation sample with the full set of fixed effects. In Panel A, the unit of observation is the firm, and in Panel B, it is the municipality; both use value-weighted aggregation based on the monetary value of commitments and cover the pre-reform period, August 1, 2014–July 30, 2015. Connected firms are those that donated to the mayor's party in the 2012 elections; unconnected donor firms donated to other parties with non-elected candidates; and unconnected non-donor firms made no campaign contributions. Liquidity is defined as (cash – accounts payable)/revenues, and municipalities are split into high- and low-liquidity groups using the 2015 median.

Table 3: Main results

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Days between verification and payment</i>					
Connected $\times$ Post	1.1 (0.74)	0.93 (0.72)	3.8*** (1.2)	2.8** (1.2)	2.7** (1.2)
Connected	-0.78 (0.68)	-0.84* (0.48)	-2.0** (0.85)	-0.07 (0.67)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0006	0.23	0.25	0.43	0.80
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	18.5	18.5	19.4	19.4	20.3
SD of dep. variable	20.7	20.7	21.2	21.2	20.4
<i>Panel B: Days between commitment and verification</i>					
Connected $\times$ Post	-1.5** (0.64)	-1.2** (0.58)	-0.68 (1.1)	0.09 (1.0)	0.11 (1.2)
Connected	-0.34 (0.65)	-2.5*** (0.52)	3.9*** (0.97)	-0.74 (0.82)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0001	0.25	0.38	0.51	0.81
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	20.2	20.2	21.8	21.8	20.4
SD of dep. variable	21.1	21.1	21.8	21.8	19.6
<i>Panel C: Log of the amount committed</i>					
Connected $\times$ Post	-0.03 (0.06)	-0.05 (0.06)	-0.13 (0.10)	-0.16 (0.10)	-0.10 (0.09)
Connected	1.3*** (0.08)	1.2*** (0.07)	1.2*** (0.10)	0.84*** (0.09)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.002	0.08	0.46	0.51	0.87
Within R ²		0.001	0.0009	0.0005	0.00
Mean dep. variable	8.1	8.1	8.2	8.2	8.6
SD of dep. variable	1.9	1.9	1.9	1.9	1.9
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: This table reports difference-in-differences estimates at the firm–municipality–time level. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is days between verification and payment (Panel A), days between commitment and verification (Panel B), or log amount committed (Panel C). Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). The main effect of Connected_{fm} is absorbed in column (5) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Days between verification and payment: Heterogeneity by liquidity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	3.2*** (1.2)	2.8** (1.3)	8.4*** (2.5)	7.2*** (2.6)	6.6** (2.7)	-0.69 (0.85)	-0.46 (0.74)	-0.81 (1.5)	-1.0 (1.3)	0.80 (1.2)
Connected	-1.9* (1.1)	-1.6** (0.81)	-3.2* (1.7)	-0.67 (1.4)		0.05 (0.83)	-0.36 (0.56)	-1.1 (1.2)	0.09 (0.88)	
Observations	338,253	338,253	214,195	214,195	94,624	409,081	409,081	273,144	273,144	124,146
Number of firms	113,693	113,693	28,832	28,832	10,167	126,554	126,554	33,777	33,777	12,657
R ²	0.0005	0.19	0.29	0.44	0.80	0.0008	0.26	0.26	0.47	0.82
Within R ²		0.00	0.00	0.00	0.0002		0.00	0.00	0.00	0.00
Mean dep. variable	21.4	21.4	22.9	22.9	24.1	16.0	16.0	17.0	17.0	17.6
SD of dep. variable	23.5	23.5	24.3	24.3	23.5	17.7	17.7	18.0	18.0	17.2
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} denotes the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Extreme delays

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Time between verification and payment above above 30 days</i>					
Connected $\times$ Post	0.006 (0.01)	0.003 (0.01)	0.04* (0.02)	0.02 (0.02)	0.03 (0.03)
Connected	-0.0005 (0.01)	-0.005 (0.01)	-0.03* (0.02)	0.009 (0.01)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0003	0.21	0.22	0.39	0.77
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	0.18	0.18	0.20	0.20	0.21
SD of dep. variable	0.39	0.39	0.40	0.40	0.40
<i>Panel B: Time between verification and payment above above 45 days</i>					
Connected $\times$ Post	0.03** (0.01)	0.02** (0.01)	0.05** (0.02)	0.03 (0.02)	0.03 (0.02)
Connected	-0.02** (0.008)	-0.02*** (0.007)	-0.03** (0.01)	-0.003 (0.01)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0003	0.15	0.19	0.32	0.73
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	0.09	0.09	0.09	0.09	0.10
SD of dep. variable	0.28	0.28	0.29	0.29	0.29
<i>Panel C: Time between verification and payment above above 60 days</i>					
Connected $\times$ Post	0.01* (0.008)	0.01 (0.008)	0.04*** (0.01)	0.03** (0.01)	0.03* (0.02)
Connected	-0.01*** (0.005)	-0.02*** (0.005)	-0.03*** (0.008)	-0.009 (0.008)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0003	0.10	0.17	0.27	0.70
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	0.05	0.05	0.05	0.05	0.05
SD of dep. variable	0.22	0.22	0.22	0.22	0.22
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: The data are at the firm-municipality-time level. Regressions take the form $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \epsilon_{fmt}$, where Connected_{fm} is a binary variable indicating if firm f donated to the winning mayor's party in municipality m in the previous election, Post_t is a binary variable indicating the post-reform period. The dependent variable is a binary variable that takes the value one when the time between verification and payment is larger than 30 days (Panel A), larger than 45 days (Panel B), or larger than 60 days (Panel C). Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in column (5) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 6: Extreme delays: Heterogeneity by liquidity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Panel A: Payment delay above 30 days</i>										
Connected × Post	0.04 (0.02)	0.03 (0.02)	0.09** (0.05)	0.08* (0.05)	0.08 (0.05)	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.03)	-0.02 (0.03)	0.02 (0.03)
Connected	-0.01 (0.02)	-0.01 (0.02)	-0.04 (0.04)	0.0007 (0.03)		0.010 (0.02)	-0.0004 (0.01)	-0.03 (0.02)	0.003 (0.02)	
Observations	338,253	338,253	214,195	214,195	94,624	409,081	409,081	273,144	273,144	124,146
Number of firms	113,693	113,693	28,832	28,832	10,167	126,554	126,554	33,777	33,777	12,657
R ²	0.0002	0.18	0.27	0.41	0.78	0.0005	0.21	0.22	0.40	0.77
Within R ²		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00
Mean dep. variable	0.24	0.24	0.26	0.26	0.28	0.14	0.14	0.15	0.15	0.15
SD of dep. variable	0.43	0.43	0.44	0.44	0.45	0.35	0.35	0.36	0.36	0.36
<i>Panel B: Payment delay above 45 days</i>										
Connected × Post	0.06*** (0.02)	0.06*** (0.02)	0.08** (0.04)	0.07* (0.04)	0.10** (0.04)	-0.002 (0.01)	-0.003 (0.01)	-0.006 (0.02)	-0.01 (0.02)	-0.004 (0.02)
Connected	-0.04** (0.01)	-0.04*** (0.01)	-0.03 (0.03)	0.0005 (0.02)		-0.004 (0.009)	-0.009 (0.008)	-0.03* (0.01)	-0.007 (0.01)	
Observations	338,253	338,253	214,195	214,195	94,624	409,081	409,081	273,144	273,144	124,146
Number of firms	113,693	113,693	28,832	28,832	10,167	126,554	126,554	33,777	33,777	12,657
R ²	0.0003	0.13	0.24	0.35	0.75	0.0003	0.15	0.20	0.33	0.73
Within R ²		0.00	0.00	0.00	0.0001		0.00	0.00	0.00	0.00
Mean dep. variable	0.12	0.12	0.13	0.13	0.14	0.06	0.06	0.06	0.06	0.06
SD of dep. variable	0.33	0.33	0.34	0.34	0.35	0.23	0.23	0.24	0.24	0.24
<i>Panel C: Payment delay above 60 days</i>										
Connected × Post	0.03* (0.01)	0.03* (0.01)	0.07** (0.03)	0.06* (0.03)	0.07* (0.04)	0.002 (0.009)	0.002 (0.008)	0.005 (0.02)	-0.0004 (0.02)	0.02 (0.02)
Connected	-0.03*** (0.010)	-0.03*** (0.010)	-0.03 (0.02)	-0.01 (0.02)		-0.006 (0.006)	-0.009* (0.005)	-0.02* (0.010)	-0.005 (0.009)	
Observations	338,253	338,253	214,195	214,195	94,624	409,081	409,081	273,144	273,144	124,146
Number of firms	113,693	113,693	28,832	28,832	10,167	126,554	126,554	33,777	33,777	12,657
R ²	0.0004	0.09	0.22	0.30	0.72	0.0003	0.11	0.19	0.29	0.70
Within R ²		0.00	0.00	0.00	0.0001		0.00	0.00	0.00	0.00
Mean dep. variable	0.07	0.07	0.08	0.08	0.08	0.03	0.03	0.03	0.03	0.03
SD of dep. variable	0.26	0.26	0.27	0.27	0.27	0.17	0.17	0.18	0.18	0.17
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. Regressions take the form $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \epsilon_{fmt}$, where Connected_{fm} is a binary variable equal to one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t is a binary variable equal to one in the post-reform period. The dependent variable is a binary variable equal to one when the time between verification and payment is greater than 30 days (Panel A), greater than 45 days (Panel B), or greater than 60 days (Panel C). Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 7: Heterogeneity by procurement method (São Paulo)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Competitive					Non-competitive				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	0.83 (1.9)	1.3 (1.8)	7.4* (4.3)	6.9 (4.3)	5.3 (3.6)	2.1* (1.2)	1.9 (1.2)	6.3** (2.5)	4.8* (2.5)	2.8 (2.3)
Connected	-4.2** (1.9)	-5.2*** (1.3)	-2.9 (3.4)	-2.7 (2.1)		-1.7 (1.1)	-0.61 (0.73)	-1.7 (1.6)	0.95 (1.2)	
Observations	97,250	97,249	80,720	80,717	38,609	270,237	270,237	188,826	188,826	97,662
Number of firms	18,000	18,000	6,592	6,592	3,308	84,691	84,691	29,577	29,577	15,499
R ²	0.003	0.34	0.16	0.47	0.80	0.002	0.22	0.30	0.45	0.77
Within R ²		0.0002	0.00	0.00	0.0001		0.00	0.00	0.00	0.00
Mean dep. variable	22.4	22.4	23.0	23.0	23.1	19.6	19.6	20.5	20.5	21.2
SD of dep. variable	22.6	22.6	22.8	22.8	21.9	21.1	21.1	21.3	21.3	20.6
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of commitments in which the supplier was selected through a *competitive* procurement method. Columns (6) to (10) use the sample of commitments in which the supplier was selected through a *non-competitive* procurement method. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} denotes the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. The analysis uses data from municipalities in the state of São Paulo, where information on procurement-method classification is linked to commitment data. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 8: Robustness checks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Analogous regression in the previous electoral cycle					Pre-reform period as the “post” period				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	-0.38 (0.79)	-0.78 (0.74)	1.6 (1.3)	0.73 (1.2)	0.94 (1.3)	-0.27 (0.57)	0.32 (0.52)	-2.0** (0.96)	-1.3 (0.92)	-1.1 (1.0)
Connected	-0.04 (1.0)	0.97 (0.73)	-0.83 (1.3)	0.79 (1.0)		-0.51 (0.58)	-1.2*** (0.42)	0.002 (0.85)	1.3* (0.71)	
Observations	426,374	426,374	301,928	301,928	142,432	744,970	744,970	538,907	538,907	261,614
Number of firms	123,067	123,067	37,637	37,637	14,382	195,543	195,543	59,363	59,363	23,772
R ²	0.001	0.28	0.25	0.47	0.82	0.0006	0.25	0.25	0.45	0.81
Within R ²		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00
Mean dep. variable	16.6	16.6	17.3	17.3	17.8	17.5	17.5	18.3	18.3	19.2
SD of dep. variable	17.1	17.1	17.4	17.4	16.4	19.0	19.0	19.4	19.4	18.6
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The table reports results from the robustness checks described in Panels (b) and (c) of Figure 1. The data are collapsed at the firm-municipality-time level. Regressions take the form $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \epsilon_{fmt}$, where y_{fmt} denotes the number of days between verification and payment. Connected_{fm} is a binary variable equal to one if firm f is connected in municipality m , and Post_t is a binary variable equal to one in the post period. Columns (1) to (5) report an analogous regression in the previous electoral cycle. In these columns, the pre period runs from August 1, 2010 to July 30, 2011 and the post period runs from October 1, 2011 to September 30, 2012. A firm f is defined as connected if it donated to the winning mayor's party in municipality m in the 2008 election. Columns (6) to (10) use the pre-reform period as the placebo post period. In these columns, the pre period runs from June 1, 2013 to May 31, 2014 and the post period runs from August 1, 2014 to July 30, 2015. A firm f is defined as connected if it donated to the winning mayor's party in municipality m in the 2012 election. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 9: Days between verification and payment: pre-trends

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected	-3.1*** (1.1)	-1.9*** (0.64)	-1.9 (1.5)	-0.59 (0.99)	-8.8* (4.8)
2013 Dummy	-3.3*** (0.25)				
2014 Dummy	-1.3*** (0.17)				
2016 Dummy	0.64*** (0.18)				
Connected × 2013 Dummy	3.0*** (0.77)	1.5** (0.66)	1.7 (1.3)	1.3 (1.2)	0.87 (1.4)
Connected × 2014 Dummy	1.9** (0.84)	2.2*** (0.68)	1.2 (1.3)	1.8 (1.1)	2.0 (1.3)
Connected × 2016 Dummy	3.6*** (1.1)	2.7** (1.1)	4.5** (1.8)	4.0** (1.8)	3.3* (1.7)
Observations	1,386,063	1,386,063	1,001,369	1,001,369	622,548
Number of firms	260,252	260,252	77,253	77,253	35,222
R ²	0.005	0.24	0.25	0.44	0.75
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	17.4	17.4	18.3	18.3	18.9
SD of dep. variable	19.0	19.0	19.4	19.4	18.9
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: The data are at the firm–municipality–year level. The outcome is days between verification and payment. We estimate the following event-study specification: $y_{fmt} = \beta_{-2} \text{Connected}_{fm} \times I_{t=2013} + \beta_{-1} \text{Connected}_{fm} \times I_{t=2014} + \beta_1 \text{Connected}_{fm} \times I_{t=2016} + \gamma \text{Connected}_{fm} + \delta_{2013} I_{t=2013} + \delta_{2014} I_{t=2014} + \delta_{2016} I_{t=2016} + \text{fixed effects} + \epsilon_{fmt}$. The omitted year is 2015. $I_{t=2013}$ is an indicator equal to one for observations in 2013, with analogous definitions for $I_{t=2014}$ and $I_{t=2016}$. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the 2012 election. Fixed effects included in each specification are indicated at the bottom of the table. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Online Appendix to "*Speed of Payment in Procurement Contracts: The Role of Political Connections*"

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A Persistence of donor relationships

The tables in this subsection document that a firm’s commercial relationship with a local government predicts the continuation of its political giving across electoral cycles. The unit of observation is a firm-municipality pair in Appendix Table A.1 and a firm-municipality-candidate or firm-municipality-party triple in Appendix Tables A.2 and A.3. The dependent variable is an indicator for whether the firm donated again in the 2012 elections (to any candidate or party in the first table, and to the *same* candidate or party in the others), conditional on having donated in 2008. The variable *local-government supplier* equals one if the firm sold to the local government during the mayoral term between the 2008 and 2012 elections; the quintile indicators refer to the amount the firm donated in 2008 (the omitted category is the first quintile). All specifications are linear probability models. As in the main analyses of the paper, we focus on the states of São Paulo, Minas Gerais, Rio Grande do Sul, and Paraná, where we observe commitments (sales) at the firm-municipality level. We also restrict the sample to firms in manufacturing and wholesale/retail trade, which are more likely to supply the products used in the main analyses: consumption material, free-distribution material, and equipment and permanent material.⁴¹

⁴¹We obtain the industry classification from the *Cadastro Nacional da Pessoa Jurídica* (CNPJ) of the Brazilian Federal Revenue Service (Receita Federal do Brasil).

Table A.1: Donating again in the same municipality

	(1)	(2)	(3)	(4)	(5)	(6)
	Candidate			Candidate or Party		
Local-government supplier	0.10*** (0.01)	0.11*** (0.01)	0.11*** (0.01)	0.10*** (0.01)	0.11*** (0.01)	0.11*** (0.01)
2nd quintile amount donated	0.02* (0.01)	0.02* (0.01)	0.02* (0.01)	0.02** (0.01)	0.02*** (0.01)	0.03*** (0.01)
3rd quintile amount donated	0.05*** (0.01)	0.05*** (0.01)	0.05*** (0.01)	0.06*** (0.01)	0.07*** (0.01)	0.07*** (0.01)
4th quintile amount donated	0.07*** (0.01)	0.08*** (0.01)	0.08*** (0.01)	0.10*** (0.01)	0.10*** (0.01)	0.11*** (0.01)
5th quintile amount donated	0.16*** (0.01)	0.16*** (0.01)	0.18*** (0.01)	0.20*** (0.01)	0.19*** (0.01)	0.21*** (0.01)
Observations	8,589	8,589	8,280	10,136	10,136	9,851
R ²	0.04	0.06	0.16	0.05	0.07	0.17
Mean dependent variable	0.13	0.13	0.13	0.15	0.15	0.15
Mean local-government supplier	0.19	0.19	0.19	0.20	0.20	0.20
Industry FE		✓	✓		✓	✓
Municipality FE			✓			✓

Notes: The unit of observation is a firm–municipality pair, and the sample is restricted to firms that donated in the municipality in the 2008 elections. The dependent variable equals one if the firm donated again in the same municipality in the 2012 elections and zero otherwise. In columns (1)–(3), donations to candidates are considered; in columns (4)–(6), donations to either candidates or parties are considered. The variable *local-government supplier* equals one if the firm sold to the municipal government. The donation-amount indicators identify the quintile of the firm’s 2008 donation, with the first quintile omitted. All specifications are estimated using linear probability models. Columns (2) and (5) include industry fixed effects, while columns (3) and (6) additionally include municipality fixed effects. Standard errors are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.2: Donating to the same candidate/party

	(1)	(2)	(3)	(4)	(5)	(6)
	Same candidate			Same candidate or party		
Local-government supplier	0.02*** (0.01)	0.03*** (0.01)	0.02*** (0.01)	0.04*** (0.01)	0.05*** (0.01)	0.04*** (0.01)
2nd quintile amount donated	0.01* (0.00)	0.01* (0.00)	0.01* (0.01)	0.03*** (0.01)	0.02*** (0.01)	0.02*** (0.01)
3rd quintile amount donated	0.03*** (0.01)	0.03*** (0.01)	0.02*** (0.01)	0.06*** (0.01)	0.05*** (0.01)	0.05*** (0.01)
4th quintile amount donated	0.07*** (0.01)	0.06*** (0.01)	0.06*** (0.01)	0.09*** (0.01)	0.07*** (0.01)	0.07*** (0.01)
5th quintile amount donated	0.08*** (0.01)	0.07*** (0.01)	0.08*** (0.01)	0.14*** (0.01)	0.12*** (0.01)	0.13*** (0.01)
Observations	11,543	11,543	11,271	11,848	11,848	11,590
R ²	0.02	0.05	0.15	0.03	0.06	0.17
Mean dependent variable	0.05	0.05	0.05	0.08	0.08	0.08
Mean local-government supplier	0.20	0.20	0.20	0.21	0.21	0.21
Industry FE		✓	✓		✓	✓
Municipality FE			✓			✓

Notes: The unit of observation is a firm–municipality–candidate or firm–municipality–party relationship, and the sample is restricted to relationships involving a donation in the 2008 elections. The dependent variable equals one if the firm donated again to the same recipient in the 2012 elections and zero otherwise. In columns (1)–(3), the outcome indicates a subsequent donation to the same candidate. In columns (4)–(6), the outcome indicates a subsequent donation to the same candidate or party. The variable *local-government supplier* equals one if the firm sold to the municipal government. The donation-amount indicators identify the quintile of the firm’s 2008 donation, with the first quintile omitted. All specifications are estimated using linear probability models. Columns (2) and (5) include industry fixed effects, while columns (3) and (6) additionally include municipality fixed effects. Standard errors are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A.3: Donating to the same candidate/party, conditional on being elected in the 2008 elections and running again in the 2012 elections

	(1)	(2)	(3)	(4)	(5)	(6)
	Same candidate			Same candidate or party		
Local-government supplier	0.05*** (0.02)	0.06*** (0.02)	0.05** (0.02)	0.04*** (0.01)	0.06*** (0.01)	0.06*** (0.02)
2nd quintile amount donated	0.02 (0.01)	0.00 (0.01)	-0.00 (0.02)	0.04*** (0.01)	0.03** (0.01)	0.01 (0.02)
3rd quintile amount donated	0.10*** (0.02)	0.07*** (0.02)	0.07*** (0.02)	0.09*** (0.01)	0.07*** (0.01)	0.07*** (0.02)
4th quintile amount donated	0.11*** (0.02)	0.08*** (0.02)	0.10*** (0.03)	0.12*** (0.02)	0.10*** (0.02)	0.10*** (0.02)
5th quintile amount donated	0.18*** (0.02)	0.15*** (0.02)	0.19*** (0.03)	0.18*** (0.02)	0.16*** (0.02)	0.16*** (0.02)
Observations	2,471	2,470	2,234	3,479	3,479	3,191
R ²	0.04	0.09	0.32	0.04	0.07	0.30
Mean dependent variable	0.13	0.13	0.13	0.13	0.13	0.13
Mean local-government supplier	0.21	0.21	0.21	0.21	0.21	0.21
Industry FE		✓	✓		✓	✓
Municipality FE			✓			✓

Notes: The unit of observation is a firm–municipality–candidate or firm–municipality–party relationship, and the sample is restricted to relationships involving a donation in the 2008 elections. For the candidate specifications, the sample is further restricted to candidates who were elected in 2008 and ran again in 2012. The dependent variable equals one if the firm donated again to the same recipient in the 2012 elections and zero otherwise. In columns (1)–(3), the outcome indicates a subsequent donation to the same candidate. In columns (4)–(6), the outcome indicates a subsequent donation to the same candidate or party. The variable *local-government supplier* equals one if the firm sold to the municipal government. The donation-amount indicators identify the quintile of the firm’s 2008 donation, with the first quintile omitted. All specifications are estimated using linear probability models. Columns (2) and (5) include industry fixed effects, while columns (3) and (6) additionally include municipality fixed effects. Standard errors are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

B Institutional setting, data and variables

B.1 Data sources, sample selection and variables

Data sources. The budget execution data originally come from State Audit Courts (*Tribunais de Contas dos Estados*, TCEs); the electoral data (including campaign contributions) come from the Superior Electoral Court (*Tribunal Superior Eleitoral*, TSE); and municipal balance-sheet, revenue, and expenditure data come from the *Sistema de Informações Contábeis e Fiscais do Setor Público Brasileiro* (SICONFI) of the Ministry of Finance. We source the data from Data Basis platform ([Dahis et al., 2022](#)).

Sample selection. The budget execution data include all municipal expenses, such as salaries, pensions, interest payments, machines, equipment, food, office material, construction, and IT services. Brazil adopts a budget classification system in which the economic classification of expenses (*elemento de despesa*) is comprised of 69 groups that are identified by a two-digit code. We restrict the data to three groups related to the purchase of goods and materials: consumption material (code 30), material for free distribution (code 32), and equipment and permanent material (code 52). The variable days between verification and payment is trimmed at the 99th percentile of the state-year distribution. The municipalities collect and treat the information and send it to the State Audit Courts yearly. As the quality of the data varies across municipalities, we exclude municipality-year pairs in which more than 80% of commitments are verified on the same day of the commitment or paid on the same day of the verification. When this happens, it suggests that the dates of the budget execution stages were incorrectly recorded. The data only include fully-executed commitments, that is, commitments that are verified and paid within the fiscal year.

Fiscal variables. There was a change in the accounting reports in 2013. Therefore, we present the variables definitions for two periods, from 2007 to 2013 and from 2014 to 2016. The definitions are such that the variables are as comparable as possible in the two periods given the information available. From 2007 to 2013, we define *cash* as the sum of cash, plus deposits in banks plus short-term financial applications (“caixa + bancos + aplicações financeiras”); *accounts payable* as expenses verified but not paid (“restos a pagar processados”); and *revenues* as current revenues (taxes, contributions, transfers from federal and state governments) minus contributions by pensioners and other deductions (“receitas correntes - contribuições sociais - deduções da receita corrente”). From 2013 to 2017, we define *cash* as cash and equivalents (“1.1.1.0.0.00.00: caixa e equivalentes de caixa”); *accounts payable* as suppliers, wages and other benefits to be paid (“2.1.1.0.0.00.00: obrigações trabalhistas, previdenciárias e assistenciais a pagar a curto prazo + 2.1.3.0.0.00.00: fornecedores e contas a pagar a curto prazo”); and *revenues* as as current revenues (taxes, contributions, transfers from federal and state

governments) minus contributions by pensioners and deductions (“1.0.0.0.00.00.00 -
receitas correntes - 1.2.1.0.00.00.00: contribuições sociais - deduções da receita).

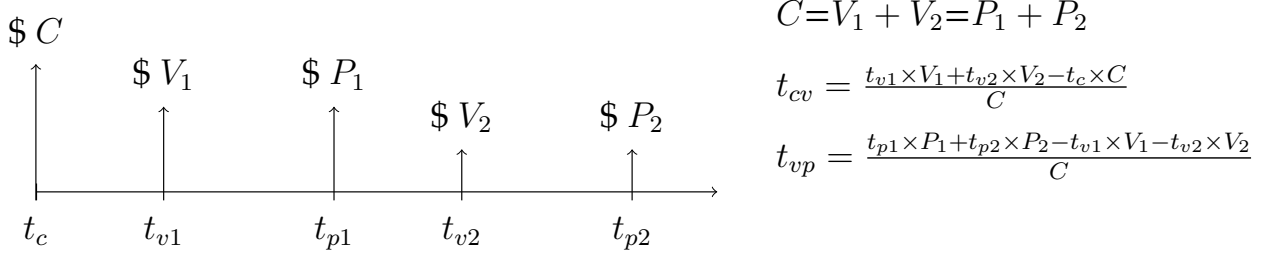
B.2 Days between verification and payment and days between commitment and verification: an example

Figure B.1: Illustration of the computation of the variables days between verification and payment (t_{vp}) and days between commitment and verification (t_{cv})

Commitment 1



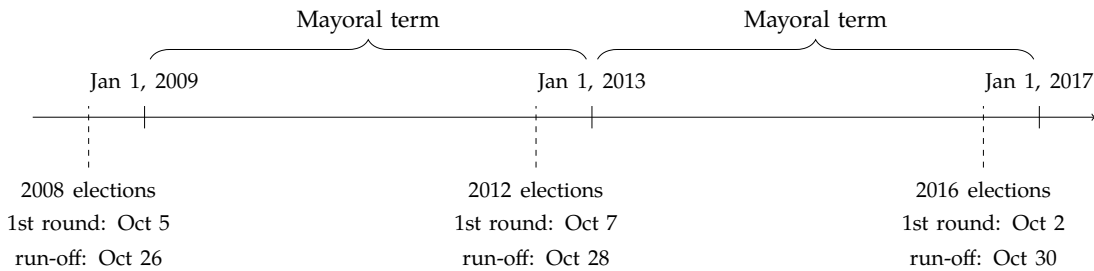
Commitment 2



Notes: Commitment 1 has value C , recorded on date t_c , one verification amount V on date t_v , and one payment amount P on date t_p . Commitment 2 also has value C , recorded on date t_c , but involves two verification stages, with amounts V_1 and V_2 recorded on dates t_{v1} and t_{v2} , respectively, and two payment amounts P_1 and P_2 recorded on dates t_{p1} and t_{p2} , respectively.

B.3 Electoral calendar: municipal elections

Figure B.2: Electoral calendar

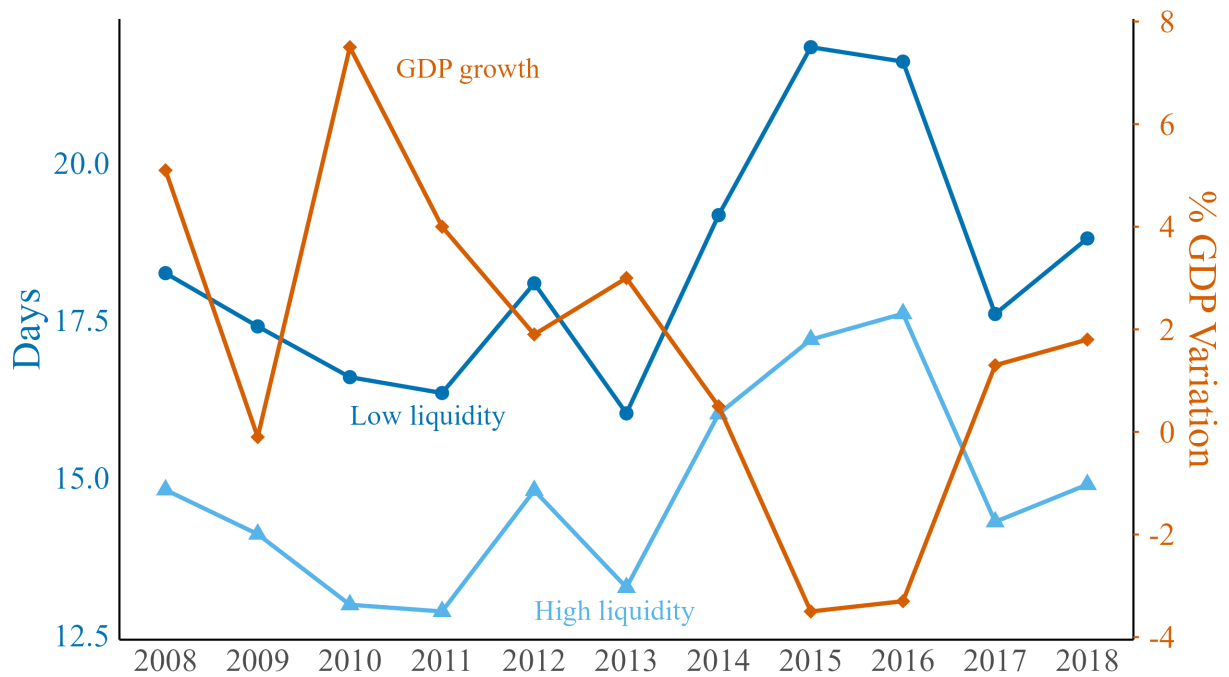


Notes: The figure shows the municipal electoral calendar for the 2008, 2012, and 2016 elections, indicating the dates of the first and second rounds and the start and end of the corresponding mayoral terms.

C Additional figures and tables

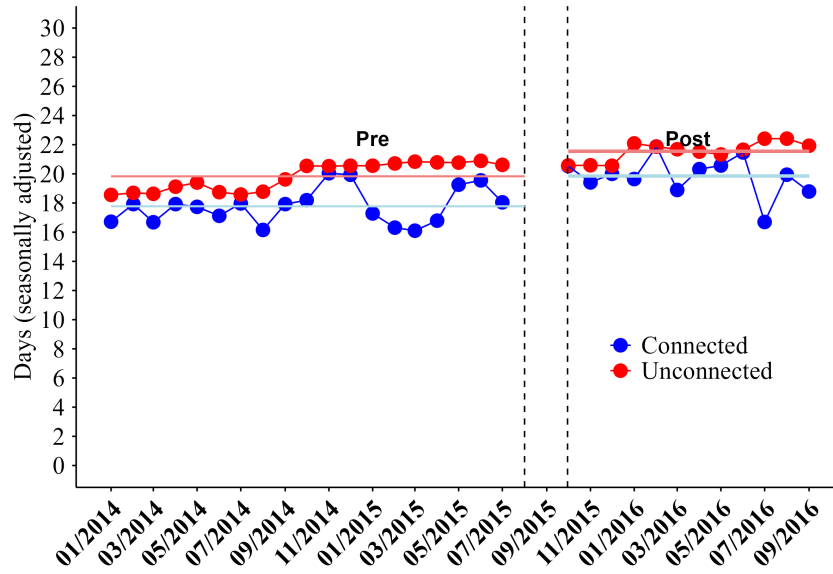
Figures

Figure C.1: Days between verification and payment: business cycle and high- versus low-liquidity variation



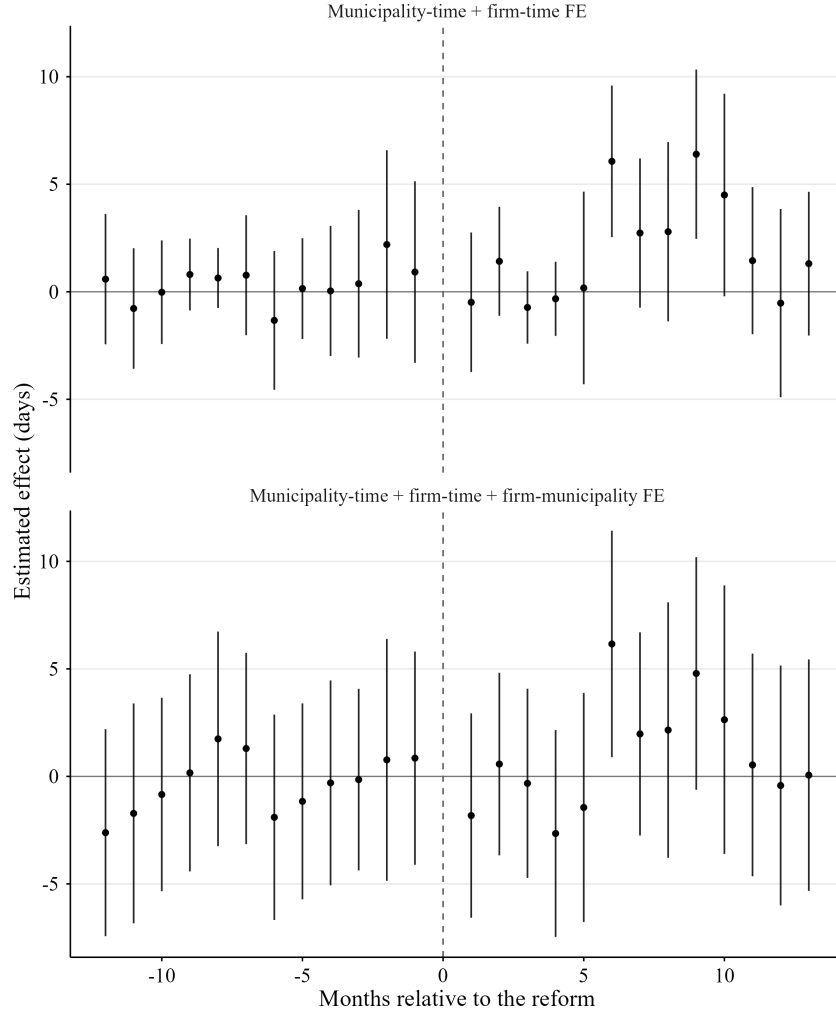
Notes: The blue lines use data at the municipality-year level and plot average days between verification and payment by liquidity group. Low-liquidity municipalities are those with liquidity below the median, and high-liquidity municipalities are those above the median. National GDP growth is plotted on the right axis.

Figure C.2: Days between verification and payment: Monthly frequency



Notes: This figure is constructed from the set of firm-municipality pairs that remain in the baseline DiD sample after removing observations dropped by the specification with municipality-time, firm-time, and firm-municipality fixed effects. For those pairs, we collapse the data to the firm-municipality-month level, where monthly payment delay is the commitment-value-weighted average of days between verification and payment across commitments within the month. Firms are classified as connected if they donated to the elected mayor's party in the 2012 municipal elections; the unconnected group pools unconnected donors and non-donors. We then seasonally adjust the monthly series by regressing payment delay on calendar-month indicators over the displayed sample window and adding the sample mean back to the residuals. The plotted points are monthly averages of this seasonally adjusted outcome by group. This version starts in January 2014 and excludes August and September 2015 from both the plotted series and the computation of pre- and post-reform averages; the horizontal segments report the average of the plotted monthly means in January 2014–July 2015 and October 2015–September 2016, respectively.

Figure C.3: Days between verification and payment: DiD monthly frequency



Notes: The data are collapsed to the firm–municipality–month level, where the outcome is the commitment-value-weighted average number of days between verification and payment within each firm–municipality–month cell. We estimate monthly event-study specifications of the form

$$y_{fmt} = \sum_{s \neq 0} \beta_s \text{Connected}_{fm} \times \mathbf{1}[t = s] + \text{FE} + \varepsilon_{fmt}$$

where y_{fmt} is payment delay for firm f in municipality m in month t , and Connected_{fm} indicates whether firm f donated to the winning mayor's party in municipality m in the previous municipal election. The omitted reference period is August 2015, so $t = 0$ corresponds to August 2015, and all coefficients are interpreted relative to that month. All months are retained in the estimation. Across panels, the fixed effects are: (i) municipality-month and firm-month fixed effects; and (ii) municipality-month, firm-month, and firm-municipality fixed effects. The coefficients β_s measure the difference in payment delays between connected and unconnected firms in event month s , relative to August 2015. Standard errors are clustered at the firm and municipality levels. Vertical bars report 95% confidence intervals.

Tables

Table C.1: Main results: Alternative thresholds for municipalities with zero delay between verification and payment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	< 50%					< 30%				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	1.2 (0.77)	1.1 (0.74)	4.3*** (1.3)	3.1** (1.3)	2.5** (1.2)	0.96 (0.78)	0.89 (0.75)	4.1*** (1.3)	2.8** (1.3)	2.7** (1.3)
Connected	-0.88 (0.70)	-0.98** (0.50)	-2.3*** (0.88)	-0.20 (0.69)		-0.73 (0.72)	-0.93* (0.51)	-2.3*** (0.89)	-0.16 (0.70)	
Observations	727,461	727,461	527,476	527,476	237,828	706,013	706,013	510,089	510,089	228,028
Number of firms	191,924	191,924	57,223	57,223	21,481	188,086	188,086	55,653	55,653	20,683
R ²	0.0004	0.22	0.25	0.43	0.80	0.0003	0.22	0.25	0.42	0.80
Within R ²		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00
Mean dep. variable	18.9	18.9	19.8	19.8	20.7	19.2	19.2	20.1	20.1	21.0
SD of dep. variable	20.7	20.7	21.2	21.2	20.5	20.8	20.8	21.2	21.2	20.4
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use municipalities for which the share of payments with zero delay between verification and payment is below 50%, while columns (6) to (10) use municipalities for which that share is below 30%. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.2: Main results: Heterogeneity by Liquidity – 30% threshold zero tvp

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	2.9** (1.3)	2.8** (1.3)	8.3*** (2.5)	7.3*** (2.7)	7.0** (2.8)	-0.70 (0.89)	-0.50 (0.78)	-0.45 (1.5)	-0.82 (1.4)	0.43 (1.3)
Connected	-2.0* (1.1)	-2.0** (0.87)	-3.7** (1.8)	-1.3 (1.4)		0.27 (0.88)	-0.30 (0.60)	-1.2 (1.2)	0.02 (0.97)	
Observations	318,255	318,255	199,864	199,864	86,244	384,700	384,700	255,194	255,194	114,224
Number of firms	108,789	108,789	27,211	27,211	9,318	120,986	120,986	32,052	32,052	11,881
R ²	0.0002	0.18	0.30	0.43	0.80	0.0006	0.25	0.27	0.46	0.81
Within R ²		0.00	0.00	0.00	0.0002		0.00	0.00	0.00	0.00
Mean dep. variable	22.2	22.2	23.7	23.7	24.9	16.7	16.7	17.6	17.6	18.3
SD of dep. variable	23.5	23.5	24.3	24.3	23.4	17.7	17.7	18.1	18.1	17.3
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.3: Main results: Heterogeneity by Liquidity – 50% threshold zero tvp

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	3.3*** (1.3)	3.1** (1.3)	9.0*** (2.5)	7.9*** (2.6)	6.3** (2.7)	-0.59 (0.87)	-0.47 (0.76)	-0.61 (1.5)	-0.93 (1.4)	0.45 (1.2)
Connected	-2.2* (1.1)	-2.0** (0.83)	-4.0** (1.8)	-1.3 (1.4)		0.14 (0.86)	-0.32 (0.58)	-1.2 (1.2)	0.04 (0.93)	
Observations	328,162	328,162	207,026	207,026	90,368	396,012	396,012	263,595	263,595	118,658
Number of firms	111,164	111,164	28,024	28,024	9,782	123,536	123,536	32,863	32,863	12,239
R ²	0.0004	0.18	0.30	0.43	0.80	0.0006	0.25	0.26	0.46	0.82
Within R ²		0.00	0.00	0.00	0.0001		0.00	0.00	0.00	0.00
Mean dep. variable	21.9	21.9	23.4	23.4	24.6	16.4	16.4	17.3	17.3	18.0
SD of dep. variable	23.5	23.5	24.3	24.3	23.4	17.7	17.7	18.1	18.1	17.3
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.4: Selection in difference-in-differences sample

		(1)	(2)
		DiD Sample	Firms that sell pre-reform only
Amount committed	Mean	35,767	11,516
	SD	206,989	66,123
	p10	480	180
	p50	5,529	1,932
	p90	64,709	19,301
Days between commitment and verification	Mean	20.44	21.04
	SD	19.58	22.70
	p10	1	0
	p50	15.10	14
	p90	46.60	52.20
Days between verification and payment	Mean	20.29	17.14
	SD	20.44	20.60
	p10	1.90	0
	p50	15	10.80
	p90	43.70	40
N		248,808	188,380

Notes: The data are at the firm-municipality-time level. There are two periods: the pre-reform period, defined as the 12-month window before the reform (August 1, 2014 to July 30, 2015), and the post-reform period, defined as the 12-month window after the reform (October 1, 2015 to September 30, 2016). Column (1) reports the sample of observations that survive the full fixed-effects specification, after dropping missing values and fixed-effect singletons. Column (2) reports firms that sell only before the reform and are therefore not used in the main estimation.

Table C.5: Firm characteristics by participation timing

	(1)	(2)	(3)
	Pre-reform only	Post-reform only	Both periods
Number of firms	51,143	48,438	96,696
Municipalities sold to (mean)	1.2	1.3	4.5
Total committed, 1,000 BRL (mean)	10	15	167
Share ever connected (%)	0.3	0.2	1.1
Days between verification and payment (mean)	14.6	15.4	17.4

Notes: This table characterizes firms by when they sell to municipal governments over the sample window. A firm is classified as selling in the pre-reform period only, the post-reform period only, or in both periods; the “Both periods” group contains the estimation sample, which is restricted further to firm-municipality pairs that transact in both periods, within firms selling to more than one municipality. For each group the table reports the number of firms and the mean number of municipalities sold to, total committed amount, share of firms ever politically connected, and number of days between verification and payment. The sample is the merged budget-execution panel before the estimation-sample restrictions, excluding Ceará and Paraíba.

Table C.6: Committed amounts in connected versus unconnected municipalities

	(1)	(2)	(3)
<i>Dependent variable: Log amount committed</i>			
Connected	1.29*** (0.082)	1.17*** (0.097)	1.17*** (0.097)
Observations	376,656	273,676	7,970
Number of firms	147,839	44,859	558
R ²	0.001	0.46	0.37
Within R ²		0.001	0.03
Mean dep. variable	8.0	8.2	8.8
Firm FE		✓	✓

Notes: This table reports descriptive regressions of the log committed amount on an indicator for the firm being politically connected in the municipality, using firm-municipality observations from the pre-reform period (August 2014–July 2015). Column (1) includes no fixed effects and recovers the across-firm gap; columns (2) and (3) include firm fixed effects and therefore compare, within a firm, the municipalities to which it is connected against those to which it is not; column (3) restricts the sample to firms connected in at least one municipality. Connected firms are those that donated to the mayor's party in the 2012 elections. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.7: Main difference-in-differences results: Full sample restrictions

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected $\times$ Post	3.7*** (1.1)	3.2*** (1.1)	3.4*** (1.2)	2.7** (1.2)	2.7** (1.2)
Connected	-4.2*** (0.95)	-2.9*** (0.63)	-2.7*** (0.99)	-0.67 (0.76)	
Observations	248,808	248,808	248,808	248,808	248,808
Number of firms	22,285	22,285	22,285	22,285	22,285
R ²	0.0004	0.31	0.27	0.49	0.80
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	20.3	20.3	20.3	20.3	20.3
SD of dep. variable	20.4	20.4	20.4	20.4	20.4
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: This table reports difference-in-differences estimates at the firm–municipality–time level. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. The sample is restricted to firm–municipality pairs with observations in both the pre- and post-reform periods, excludes observations from Ceará and Paraíba, and includes only firms that sell to more than one municipality. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). The main effect of Connected_{fm} is absorbed in column (5) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.8: Main difference-in-differences results: Coalition

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected coalition $\times$ Post	0.91 (0.66)	0.76 (0.63)	3.3*** (1.1)	2.3** (1.1)	1.7 (1.1)
Connected coalition	-1.0* (0.61)	-1.3*** (0.43)	-2.0** (0.78)	-0.31 (0.63)	
Observations	750,621	750,621	545,891	545,891	248,808
Number of firms	196,277	196,277	58,876	58,876	22,285
R ²	0.0006	0.23	0.25	0.43	0.80
Within R ²		0.00	0.00	0.00	0.00
Mean dep. variable	18.5	18.5	19.4	19.4	20.3
SD of dep. variable	20.7	20.7	21.2	21.2	20.4
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: This table reports difference-in-differences estimates at the firm–municipality–time level. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor’s coalition in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). The main effect of Connected_{fm} is absorbed in column (5) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.9: Main difference-in-differences results: Coalition - Heterogeneity by Liquidity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Dependent variable: Days between verification and payment</i>										
Connected coalition × Post	2.3** (1.1)	2.1* (1.2)	7.6*** (2.1)	6.7*** (2.3)	6.2** (2.5)	-0.26 (0.71)	-0.31 (0.62)	-0.94 (1.3)	-1.5 (1.2)	-0.27 (1.2)
Connected coalition	-1.8* (0.99)	-2.2*** (0.75)	-2.6* (1.5)	-0.77 (1.3)		-0.46 (0.73)	-0.65 (0.47)	-0.91 (1.1)	0.03 (0.80)	
Observations	338,253	338,253	214,195	214,195	94,624	409,081	409,081	273,144	273,144	124,146
Number of firms	113,693	113,693	28,832	28,832	10,167	126,554	126,554	33,777	33,777	12,657
R ²	0.0005	0.19	0.29	0.44	0.80	0.0008	0.26	0.26	0.47	0.82
Within R ²		0.00	0.00	0.00	0.0002		0.00	0.00	0.00	0.00
Mean dep. variable	21.4	21.4	22.9	22.9	24.1	16.0	16.0	17.0	17.0	17.6
SD of dep. variable	23.5	23.5	24.3	24.3	23.5	17.7	17.7	18.0	18.0	17.2
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's coalition in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.10: Main results (São Paulo)

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected $\times$ Post	2.8** (1.3)	2.3* (1.3)	7.1*** (2.3)	5.6** (2.3)	4.9** (2.3)
Connected	-2.2* (1.2)	-1.3* (0.74)	-2.6 (1.6)	-0.58 (1.1)	
Observations	318,107	318,107	228,621	228,621	105,372
Number of firms	88,375	88,375	27,756	27,756	10,522
R ²	0.002	0.23	0.24	0.43	0.81
Within R ²		0.00	0.00	0.00	0.0001
Mean dep. variable	20.2	20.2	21.2	21.2	22.3
SD of dep. variable	21.3	21.3	21.7	21.7	20.9
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: The data are at the firm-municipality-time level, restricted to the state of São Paulo. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \text{FE} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Column (1) includes no fixed effects, column (2) includes municipality-time fixed effects, columns (3) and (4) restrict the sample to firms that sold in at least two different municipalities before and after the reform and include firm-time fixed effects, with column (4) also including municipality-time fixed effects. Column (5) adds firm-municipality. Connected_{fm} is a binary variable indicating if firm f donated to the winning mayor's party in municipality m in the previous election, Post_t is a binary variable indicating the post-reform period. Standard errors clustered at the firm and municipality levels are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.11: Heterogeneity by liquidity (São Paulo)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Low liquidity					High liquidity				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	6.5*** (1.8)	6.3*** (2.1)	12.8*** (4.1)	11.4** (4.6)	10.0** (4.8)	-0.96 (1.6)	-1.0 (1.3)	2.2 (2.8)	2.2 (2.3)	4.3** (2.0)
Connected	-3.2 (2.0)	-3.3** (1.3)	-4.8 (3.3)	-2.9 (2.4)		-1.2 (1.3)	0.36 (0.83)	-2.8 (2.2)	-0.40 (1.6)	
Observations	144,144	144,144	88,890	88,890	39,574	173,963	173,963	112,404	112,404	51,424
Number of firms	51,540	51,540	13,422	13,422	4,688	58,189	58,189	15,968	15,968	5,847
R ²	0.002	0.19	0.28	0.44	0.80	0.004	0.26	0.26	0.48	0.82
Within R ²		0.00	0.0001	0.0001	0.0004		0.00	0.00	0.00	0.0001
Mean dep. variable	23.9	23.9	25.5	25.5	27.1	17.1	17.1	18.1	18.1	18.9
SD of dep. variable	24.6	24.6	25.2	25.2	24.4	17.6	17.6	17.7	17.7	16.9
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: The data are at the firm-municipality-time level. Columns (1) to (5) use the sample of municipalities in the *low-liquidity* group, defined as municipalities with liquidity at or below the median liquidity in 2015. Columns (6) to (10) use the sample of municipalities in the *high-liquidity* group, defined as municipalities with liquidity above the median liquidity in 2015. The estimated equation is $y_{fmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table. The main effect of Connected_{fm} is absorbed in columns (5) and (10) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.12: Main results (São Paulo, commitment level)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Dependent variable: Days between verification and payment</i>								
Connected $\times$ Post	2.6** (1.2)	2.2* (1.2)	4.9** (2.1)	4.4** (2.1)	3.5* (2.0)	3.5* (2.0)	3.6* (2.0)	3.6* (2.0)
Connected	-2.2* (1.2)	-1.4* (0.78)	-2.0 (1.7)	-0.03 (1.0)				
Observations	2,732,353	2,732,353	2,702,575	2,702,575	2,654,061	2,654,059	2,654,061	2,654,059
Number of firms	82,129	82,129	62,052	62,052	59,763	59,763	59,763	59,763
R ²	0.003	0.20	0.29	0.42	0.66	0.66	0.66	0.66
Within R ²	0.001	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean dep. variable	23.6	23.6	23.7	23.7	23.7	23.7	23.7	23.7
SD of dep. variable	23.9	23.9	23.9	23.9	23.9	23.9	23.9	23.9
Committed-value-bin FE	✓	✓	✓	✓	✓	✓	✓	✓
Municipality-time FE		✓		✓	✓	✓	✓	✓
Firm-time FE			✓	✓	✓	✓	✓	✓
Firm-municipality FE					✓	✓	✓	✓
Subcategory-time FE						✓		✓
Proc. method-time FE							✓	✓

Notes: This table reports difference-in-differences estimates using commitment-level data for the state of São Paulo. The dependent variable is the number of days between verification and payment. The most saturated estimating equation is $y_{cfmt} = \beta \text{Connected}_{fm} \times \text{Post}_t + \gamma \text{Connected}_{fm} + \alpha_{b(c)} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \alpha_{st} + \alpha_{pt} + \varepsilon_{cfmt}$, where c indexes commitments and $\alpha_{b(c)}$ denotes committed-value-bin fixed effects. The bins are defined by 100 quantile groups of committed value constructed from the pooled pre- and post-reform estimation sample. Connected_{fm} equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table: committed-value-bin FE, municipality-time FE, firm-time FE, firm-municipality FE, subcategory-time FE, and procurement-method $\times$ time FE. The main effect of Connected_{fm} is absorbed in columns (5)–(8) by firm-municipality fixed effects. The sample is restricted to commitment-level observations in São Paulo with non-missing and non-zero days between verification and payment and to firm-municipality-post cells present in the aggregate São Paulo DiD sample. Regressions are weighted by each commitment's share of total committed value within the firm-municipality-time cell. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.13: Descriptive statistics for commitments: pre-reform period

	(1)	(2)	(3)
	Full sample	Large (>p50)	Small (<p50)
Amount committed (BRL)	2,622	5,079	165
Days between commitment and payment	37.4	38.8	35.9
Days between commitment and verification	17.3	18.8	15.7
Days between verification and payment	20.1	19.9	20.2
% going to connected firms	1.3	1.4	1.3
Observations	3,196,382	1,598,189	1,598,183

Notes: This table presents descriptive statistics at the commitment level for the pre-reform period, which spans from August 1, 2014, to July 30, 2015. The sample is restricted to commitments satisfying the sample construction criteria described in Section 3.2. Column (1) reports the full pre-reform sample. Columns (2) and (3) split commitments according to whether the committed amount is above or below the median committed amount in the pre-reform sample. Connected firms are those that donated to the mayor's party, that is, the party of the candidate elected in the 2012 mayoral elections.

Table C.14: Donation selection and municipality characteristics

	(1)	(2)
	Donated in 2012	
	Full sample	Estimation sample
Rigidity	-0.002 (0.002)	-0.008** (0.004)
Liquidity	-0.0003 (0.0004)	0.0001 (0.001)
Winner's margin of victory	0.0002 (0.0005)	-0.0002 (0.001)
Incumbent party	0.0004* (0.0002)	-0.0003 (0.0005)
Runoff election	0.000 (0.0010)	0.002 (0.002)
Log population	0.003*** (0.0001)	0.005*** (0.0003)
Log GDP per capita	-0.0002 (0.0002)	-0.0007 (0.0005)
Observations	379,366	108,957
Number of firms	57,820	19,917
R ²	0.300	0.323
Within R ²	0.004	0.006
Mean dep. variable	0.004	0.006
Firm FE	✓	✓

Notes: This table studies whether firms are more likely to donate in municipalities with different pre-determined municipal characteristics. The unit of observation is a firm-municipality pair. The dependent variable is an indicator equal to one if firm f donated to any mayoral candidate in municipality m in the 2012 municipal election, and zero otherwise. Both columns report linear probability models estimated by OLS. Column (1) uses firm-municipality pairs in the full sample, while column (2) uses firm-municipality pairs in the estimation sample. Both specifications include firm fixed effects, so coefficients are identified from within-firm variation across municipalities. Municipality characteristics are measured in 2012. Rigidity is measured in 2012 and is defined as the wage bill divided by current revenues. Liquidity is measured in 2012 and is defined as cash minus accounts payable, divided by current revenues. The winner's margin of victory is measured in the 2012 mayoral election and is the vote-share margin between the winning candidate and the relevant challenger. Incumbent party is an indicator for whether the incumbent party won the 2012 mayoral election. Runoff election is an indicator for municipalities with a second-round mayoral election in 2012. Population and GDP per capita are measured in 2012 and enter in logs. Standard errors are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.15: Probability of being paid after 30, 45, or 60 days

		(1)	(2)	(3)	(4)
		Full sample	Estimation sample	Estimation sample - low liquidity	Estimation sample - high liquidity
30+ days	Mean	0.18	0.21	0.27	0.15
45+ days	Mean	0.09	0.10	0.14	0.06
60+ days	Mean	0.05	0.05	0.08	0.03
Observations		750,621	248,808	109,962	137,836

Notes: The data are at the firm-municipality-time level. The periods are two: the pre-reform period, which is the 12-month window before the reform (August 1, 2014 to July 30, 2015), and the post-reform period, which is the 12-month window after the reform (October 1, 2015 to September 30, 2016). Column (1) refers to the full sample. Column (2) refers to the sample used to estimate the difference-in-differences model, defined as the observations that survive the full fixed-effects specification after dropping missing values and fixed-effect singletons; this sample includes only firms that sold to at least two municipalities and sold in both the pre- and post-reform periods. Columns (3) and (4) split the sample of column (2) into two according to the median of the municipal liquidity measure as of December 2015.

Table C.16: Descriptive statistics for firms and municipalities: heterogeneity by procurement method (São Paulo)

	(1)	(2)	(3)	(4)	(5)	(6)
	Competitive			Non-competitive		
	Connected	Unconnected		Connected	Unconnected	
		Donor	Non-donor		Donor	Non-donor
<i>Panel A: Firms</i>						
Days between commitment and payment	48.1 (29.2)	42.8 (25.9)	46.5 (30.1)	39.8 (27.9)	38.2 (27.3)	38.2 (27.5)
Days between commitment and verification	25.8 (22.6)	16.4 (18.9)	22.4 (21.4)	17.9 (17.5)	15.9 (17.8)	14.3 (17.3)
Days between verification and payment	22.3 (24.5)	26.4 (22.6)	24.1 (24.3)	22 (23.3)	22.3 (23.8)	23.9 (23.3)
Number of municipalities firm sells to	8.6 (25)	3.3 (6.9)	3.4 (10.1)	5 (11)	3 (7.3)	2.3 (4.5)
Total committed (1,000 BRL)	2,226 (4,883)	382.3 (1,121)	357.7 (1,789)	147.9 (699.9)	29.4 (57.7)	47.4 (860.6)
Number of municipalities firm donates to	1.2 (0.6)	1 (0.1)	- -	1.1 (0.5)	1 (0.2)	- -
Total donated to losers (BRL)	38,061 (154,407)	94,384 (284,584)	- -	99,076 (1,007,917)	52,272 (192,066)	- -
Total donated to winners (BRL)	311,452 (666,015)	- -	- -	251,625 (987,386)	- -	- -
Number of connections	35.9 (94)	- -	- -	18.5 (43.5)	- -	- -
Number of firms	79	95	6,451	164	252	19,261
Number of firm-municipalities	2,835	3,052	287,173	3,038	4,359	408,886

Notes: This table presents the mean and standard deviation (in parentheses) of firm and municipality characteristics restricted to São Paulo. The firms are those that enter our estimation sample. The data are at the firm level for the pre-reform period. The aggregation uses the monetary values of the operations as weights. The pre-reform period spans from August 1, 2014, to July 30, 2015. Connected firms are those that donated to the mayor's party (that is, the party of the candidate that was elected) in the 2012 elections; unconnected-donor firms are those that donated to the party of candidates that were not elected in the 2012 elections; unconnected-non-donor firms are those that do not make campaign contributions. Reported donations correspond to the 2012 mayoral elections.

Table C.17: Days between verification and payment: DiD monthly frequency –
Observations

(1)	(2)	(3)
Month relative to reform	Municipality-time + firm-time FE	Municipality-time + firm-time + firm-municipality FE
-12	54,286	43,487
-11	53,555	44,237
-10	48,137	40,509
-9	36,981	31,884
-8	16,556	13,954
-7	51,050	43,709
-6	50,651	43,715
-5	62,330	52,688
-4	55,011	47,642
-3	56,655	48,921
-2	55,373	48,045
-1	53,604	46,860
1	42,602	41,632
2	42,309	37,742
3	35,355	31,490
4	15,934	13,932
5	50,915	44,791
6	55,582	48,482
7	64,185	54,701
8	57,913	49,994
9	58,738	50,458
10	60,944	51,616
11	55,928	47,317
12	57,988	48,532

Notes: This table plots the number of observations by month used in the monthly DiD exercise.

Table C.18: Main DiD: Heterogeneity by term

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected $\times$ Post $\times$ First-term mayor	-0.37 (1.6)	0.30 (1.6)	-0.88 (2.6)	0.34 (2.6)	-0.56 (2.4)
Connected $\times$ Post	1.4 (1.4)	0.72 (1.4)	4.6** (2.3)	2.7 (2.2)	2.9 (1.8)
Connected $\times$ First-term mayor	1.0 (1.6)	-0.47 (0.94)	-1.1 (1.9)	-1.5 (1.4)	
Post $\times$ First-term mayor	0.34 (0.38)		0.81** (0.37)		
Connected	-1.3 (1.4)	-0.46 (0.69)	-1.4 (1.6)	0.92 (1.0)	
Post	0.69** (0.31)				
First-term mayor	1.3** (0.62)		0.54 (0.56)		
Observations	724,160	724,160	523,934	523,934	238,432
Number of firms	191,554	191,554	57,054	57,054	21,506
R ²	0.002	0.23	0.25	0.43	0.80
Within R ²		0.00	0.0005	0.00	0.00
Mean dep. variable	18.6	18.6	19.5	19.5	20.4
SD of dep. variable	20.7	20.7	21.2	21.2	20.5
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: This table reports difference-in-differences estimates at the firm–municipality–time level, allowing the effect of political connections to vary with the mayor’s term status. The estimated equation is $y_{fmt} = \beta_1 (Connected_{fm} \times Post_t \times FirstTerm_m) + \beta_2 (Connected_{fm} \times Post_t) + \gamma_1 (Connected_{fm} \times FirstTerm_m) + \gamma_2 Connected_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. $Connected_{fm}$ equals one if firm f donated to the winning mayor’s party in municipality m in the previous election, and $Post_t$ equals one in the post-reform period. $FirstTerm_m$ equals one if the mayor in municipality m was serving a first term; second-term mayors are the omitted reference group. Thus, β_1 captures the differential connected-firm post-reform effect in first-term municipalities relative to second-term municipalities, while β_2 captures the corresponding effect for second-term municipalities. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). In column (5), the terms $Connected_{fm} \times FirstTerm_m$ and $Connected_{fm}$ are absorbed by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.19: Main results: RDD – All firms

	(1)	(2)	(3)	(4)	(5)	(6)
	All firms			Connected firms only		
<i>Dependent variable: Days between verification and payment</i>						
First-term mayor	0.741 (1.278)	0.491 (1.735)	-0.026 (0.998)	7.588 (5.035)	0.032 (6.352)	16.581*** (4.124)
Robust 90% CI	[-4.346 ; 2.858]	[-2.486 ; 7.264]	[-2.011 ; 3.310]	[-11.261 ; 16.654]	[-10.491 ; 21.439]	[5.210 ; 28.342]
Kernel Type	Triangular	Triangular	Triangular	Triangular	Triangular	Triangular
BW Type	CCT	.5CCT	2CCT	CCT	.5CCT	2CCT
BW	0.135	0.067	0.269	0.161	0.081	0.322
Observations	688	688	688	214	214	214

Notes: This table reports coefficients from a regression discontinuity (RD) specification that adapts the framework of [Dahis et al. \(2025\)](#) to assess heterogeneity in the difference-in-differences estimates by mayor term status. The dependent variables are municipality-level changes in the days between verification and payment between the pre- and post-reform periods. For each municipality, outcomes are first averaged separately in the pre- and post-reform periods, and the RD is estimated using the difference between these two averages as the outcome. Columns (1)–(3) use all firms in the municipality, while columns (4)–(6) restrict the underlying averages to politically connected firms only. A firm is defined as politically connected if it donated to the winning mayor’s party in the previous election. The running variable is the first-term candidate’s margin of victory, defined as the vote share of the strongest challenger minus the vote share of the incumbent mayor, so that it is positive in municipalities where the incumbent was defeated and a first-term mayor was elected, and negative where the incumbent was reelected to a second term. The sample is restricted to municipalities in which an incumbent candidate was present in the election and the incumbent and challenger were the top two candidates. All specifications include controls for runoff elections, party incumbency, and state fixed effects. Estimates are obtained using local linear RD regressions with a triangular kernel. For each sample, the reported columns use the CCT bandwidth, half the CCT bandwidth, and twice the CCT bandwidth, respectively, where CCT denotes the MSE-optimal bandwidth of Calonico et al. (2014). The row “Robust 90% CI” reports robust bias-corrected confidence intervals from `rdrobust`. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.20: Main DiD - Placebo: Heterogeneity by term

	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Days between verification and payment</i>					
Connected $\times$ Post $\times$ First-term mayor	0.35 (1.7)	-0.77 (1.6)	2.0 (2.7)	1.4 (2.4)	0.49 (2.7)
Connected $\times$ Post	-0.65 (1.3)	-0.35 (1.2)	0.31 (2.0)	-0.24 (1.7)	0.40 (2.2)
Connected $\times$ First-term mayor	1.3 (2.0)	-0.03 (1.5)	-0.44 (2.5)	-1.1 (2.0)	
Post $\times$ First-term mayor	0.78* (0.44)		1.0** (0.44)		
Connected	-0.69 (1.4)	1.1 (1.2)	-0.45 (1.7)	1.5 (1.5)	
Post	0.67** (0.29)				
First-term mayor	0.21 (0.73)		0.31 (0.64)		
Observations	413,223	413,223	290,808	290,807	137,042
Number of firms	120,721	120,721	36,591	36,591	13,948
R ²	0.002	0.28	0.25	0.47	0.83
Within R ²		0.00	0.0008	0.00	0.00
Mean dep. variable	16.7	16.7	17.4	17.4	17.9
SD of dep. variable	17.2	17.2	17.4	17.4	16.5
Municipality-time FE		✓		✓	✓
Firm-time FE			✓	✓	✓
Firm-municipality FE					✓

Notes: This table reports placebo difference-in-differences estimates at the firm–municipality–time level, using the previous electoral cycle. The placebo sample compares the 2010–2011 pre-period with the 2011–2012 post-period, and political connections are defined using donations to the winning mayor’s party in the 2008 election. The estimated equation is $y_{fmt} = \beta_1 (\text{Connected}_{fm} \times \text{Post}_t \times \text{FirstTerm}_m) + \beta_2 (\text{Connected}_{fm} \times \text{Post}_t) + \gamma_1 (\text{Connected}_{fm} \times \text{FirstTerm}_m) + \gamma_2 \text{Connected}_{fm} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{fmt}$, where y_{fmt} is the number of days between verification and payment. Connected_{fm} equals one if firm f donated to the winning mayor’s party in municipality m in the 2008 election, and Post_t equals one in the placebo post- period. FirstTerm_m equals one if the mayor elected in municipality m in 2008 was serving a first term; second-term mayors are the omitted reference group. Thus, β_1 captures the differential connected-firm placebo post-period effect in first-term municipalities relative to second- term municipalities, while β_2 captures the corresponding effect for second-term municipalities. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). In column (5), the terms $\text{Connected}_{fm} \times \text{FirstTerm}_m$ and Connected_{fm} are absorbed by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table C.21: Main results: Services and construction

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Service					Construction				
<i>Dependent variable: Days between verification and payment</i>										
Connected × Post	0.95 (0.75)	0.96 (0.69)	0.28 (1.3)	0.21 (1.2)	-0.36 (1.3)	-2.7 (2.1)	-2.3 (2.2)	-2.2 (2.8)	-1.5 (3.6)	-8.0 (8.4)
Connected	1.7** (0.66)	1.4*** (0.50)	-1.1 (0.98)	-0.27 (0.87)		2.4 (1.9)	2.3 (1.9)	0.60 (2.0)	2.1 (2.2)	
Observations	698,720	698,720	426,917	426,917	205,792	30,093	29,675	13,203	11,950	1,974
Number of firms	247,578	247,578	58,763	58,763	18,869	16,146	15,991	3,079	2,834	372
R ²	0.0006	0.17	0.37	0.46	0.84	0.0002	0.29	0.40	0.66	0.93
Within R ²		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.003
Mean dep. variable	13.7	13.7	13.5	13.5	13.4	13.0	13.0	12.6	12.8	12.8
SD of dep. variable	18.1	18.1	18.3	18.3	17.2	17.8	17.8	17.5	17.6	16.7
Municipality-time FE		✓		✓	✓		✓		✓	✓
Firm-time FE			✓	✓	✓			✓	✓	✓
Firm-municipality FE					✓					✓

Notes: This table reports difference-in-differences estimates at the firm–municipality–time level. The estimated equation is $y_{f,mt} = \beta \text{Connected}_{f,m} \times \text{Post}_t + \gamma \text{Connected}_{f,m} + \alpha_{mt} + \alpha_{ft} + \alpha_{fm} + \varepsilon_{f,mt}$, where $y_{f,mt}$ is the number of days between verification and payment. $\text{Connected}_{f,m}$ equals one if firm f donated to the winning mayor's party in municipality m in the previous election, and Post_t equals one in the post-reform period. Fixed effects included in each specification are indicated at the bottom of the table: municipality-time FE (cols. 2, 4, 5), firm-time FE (cols. 3, 4, 5), and firm-municipality FE (col. 5). The main effect of $\text{Connected}_{f,m}$ is absorbed in column (5) by firm-municipality fixed effects. Standard errors (in parentheses) are two-way clustered at the firm and municipality levels. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.